

A Survey of Modulation Classification Using Deep Learning: Signal Representation and Data Preprocessing

Shengliang Peng^{ID}, *Member, IEEE*, Shujun Sun^{ID}, and Yu-Dong Yao^{ID}, *Fellow, IEEE*

Abstract—Modulation classification is one of the key tasks for communications systems monitoring, management, and control for addressing technical issues, including spectrum awareness, adaptive transmissions, and interference avoidance. Recently, deep learning (DL)-based modulation classification has attracted significant attention due to its superiority in feature extraction and classification accuracy. In DL-based modulation classification, one major challenge is to preprocess a received signal and represent it in a proper format before feeding the signal into deep neural networks. This article provides a comprehensive survey of the state-of-the-art DL-based modulation classification algorithms, especially the techniques of signal representation and data preprocessing utilized in these algorithms. Since a received signal can be represented by either features, images, sequences, or a combination of them, existing algorithms of DL-based modulation classification can be categorized into four groups and are reviewed accordingly in this article. Furthermore, the advantages as well as disadvantages of each signal representation method are summarized and discussed.

Index Terms—Deep learning (DL), feature representation, image representation, modulation classification, sequence representation.

I. INTRODUCTION

FULL radio spectrum awareness is crucial in order to achieve efficient, reliable, and secure communications for future wireless communications systems and it includes many identification and classification-related technical tasks, such as spectrum sensing [1], radio coverage determination [2], fading channel identification [3], interference environment identification [4], channel code recognition [5], modulation classification [6]–[10], medium access control (MAC) protocol classification [11]–[13], user terminal identification [14], and cellular system identification [15]–[17]. This article addresses the topic of modulation classification and, specifically, focuses on the signal representation and data preprocessing aspect in modulation classification.

Manuscript received 10 November 2020; revised 13 March 2021; accepted 26 May 2021. Date of publication 14 June 2021; date of current version 1 December 2022. This work was supported by the Fundamental Research Funds for the Central Universities under Grant ZQN-708. (Corresponding author: Yu-Dong Yao.)

Shengliang Peng and Shujun Sun are with the College of Information Science and Technology, Huaqiao University, Xiamen 361021, China.

Yu-Dong Yao is with the Department of Electrical and Computer Engineering, Stevens Institute of Technology, Hoboken, NJ 07030 USA (e-mail: yyao@stevens.edu).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TNNLS.2021.3085433>.

Digital Object Identifier 10.1109/TNNLS.2021.3085433

Modulation classification is broadly applied in both civilian and military fields. In cognitive radio, it is essential for the full awareness of a wireless environment. In spectrum surveillance, modulation information is important to determine whether radio regulations are followed. In electronic warfare, it is necessary to identify the modulation schemes of enemy signals when implementing radio jamming and deception. Moreover, modulation classification is regarded as one of the key techniques for communications systems with adaptive modulation capabilities.

There have been significant efforts in the research of modulation classification. Earlier work can be summarized into two categories, namely likelihood-based (LB) and feature-based (FB) methods [6]. The former, which treats modulation classification as a multiple hypothesis testing problem, leads to optimal solutions but suffers from high computation complexity [18]. The latter, which extracts features to represent the signal, is able to achieve near-optimal performance with reduced complexity if proper features and classifiers are adopted [6]. In [19], two approximate LB algorithms using the Gauss–Legendre and Gauss–Hermite quadrature rules are proposed to classify linearly modulated signals with reduced complexity. In [20], a robust FB algorithm that applies higher order cumulants (HOCs) as features is presented to combat multipath fading. Cyclostationarity features and approximate entropy features are exploited for modulation classification in [21] and [22], respectively. In [23], Kullback–Leibler divergence is suggested to identify multilevel amplitude phase shift keying (APSK) signals. In [24], the fourth-order spectrum features as well as compressive sensing are utilized to discriminate similar constellations. Over multiple-input-multiple-output (MIMO) channels, HOC features and artificial neural network (ANN) classifiers are the best choices for the tradeoff between performance and complexity [25]. In MIMO networks, a decision tree classifier is more efficient than an ANN classifier with reasonable training time [26]. Other classifiers, such as support vector machine (SVM) and K-nearest neighbor (KNN), are also exploited by FB methods in [24], [25], and [27].

In recent years, with the flourishing of deep learning (DL), DL-based modulation classification is becoming more and more popular [28], [29]. There are several advantages of implementing DL for modulation classification [10]. First, massive amount of data required in DL can be easily obtained

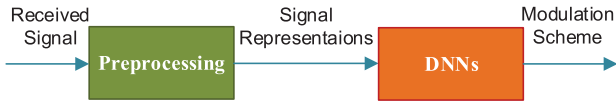


Fig. 1. Two stages in DL-based modulation classification.

in communications systems. Second, manual feature selections, which can be a significant challenge in modulation classification, are not needed in DL. Third, with the rapid and significant advancement of DL technologies, considerable potentials can be found for modulation classification under complex scenarios.

DL-based modulation classification usually comprises two successive stages. In the first stage, a received signal is pre-processed for proper representation for later use. In the second stage, deep neural networks (DNNs) are utilized to process the signal representations and determine the modulation scheme. The first stage is time-consuming but conducted off-line beforehand, whereas the second stage is performed online repeatedly but costs far less time, which facilitates the use of DL-based modulation classification in future high-speed real-time applications. This article gives a survey of DL-based modulation classification, especially the techniques of signal representation and data preprocessing. Since the received signal can be represented by features, images, sequences, or their combinations before being fed into DNNs, existing DL-based modulation classification algorithms can be categorized into the four groups accordingly.

Some of the highlights of this survey paper are listed as follows.

- 1) This article provides a comprehensive survey on the recent advancements in DL-based modulation classification from the perspective of signal representation and data preprocessing, which emphasizes the important step between the DL tool and the modulation classification task.
- 2) This article categorizes the existing DL-based modulation classification algorithms according to the techniques of signal representation into four groups, including feature representation, image representation, sequence representation, and combined representation.
- 3) This article gives detailed examples of each group of signal representation that can serve as a starting point for further investigation of DL-based modulation classification.
- 4) This article discusses both the advantages and disadvantages of each signal representation technique, which are essential for designing future studies on DL-based modulation classification.

The rest of this article is organized as follows. Section II formulates the problem of modulation classification. Section III presents DL-based modulation classification methods with feature representation. Section IV explains DL methods with image representation. Section V introduces DL methods that adopt sequences for signal representation. Section VI shows how to exploit the combined representation for modulation

classification. Summary and discussions of various DL methods are given in Section VII. Finally, Section VIII concludes this article.

II. PROBLEM FORMULATION

Consider a received signal with a complex envelope [30]

$$y(k) = Ae^{j(2\pi f_0 kT + \theta_k)} \cdot \sum_{l=-\infty}^{\infty} x(l)h(kT - lT + \epsilon_T T) + g(k) \quad (1)$$

where $x(l)$ represents the symbol sequence, A is an unknown amplitude factor, f_0 denotes the carrier frequency offset, θ_k is the phase jitter, T represents the symbol spacing, $h(\cdot)$ denotes the residual channel effect, ϵ_T is the time error, and $g(k)$ represents the additive Gaussian noise. The task of modulation classification is to determine which modulation scheme has been utilized by $y(k)$ from a candidate modulation set.

According to the DL-based modulation classification, the received signal is first preprocessed for signal representation and then fed into DNNs to infer the modulation scheme, as shown in Fig. 1. Various techniques can be applied in the stage of data preprocessing and the received signal is represented in various formats, such as features, images, sequences, and their combinations.

III. DL METHODS: FEATURE REPRESENTATION

The most direct approaches to utilize DL for modulation classification are DL methods with feature representation. In the first stage, some features are extracted from the received signal. In the second stage, DNNs are adopted instead of classical classifiers (e.g., SVM and KNN) to learn from the extracted features and determine the modulation scheme.

A. Cumulants Features

Cumulants are the most prevalent features used to classify modulation schemes. The n th-order cumulant with m conjugate variables is given by

$$C_{nm} = \text{cum} \left(\underbrace{y(k), \dots, y(k)}_{n-m}, \underbrace{y^*(k), \dots, y^*(k)}_m \right). \quad (2)$$

Some cumulants of zero-mean signals can be calculated with moments as follows:

$$\begin{aligned} C_{20} &= \text{cum}(y(k), y(k)) = M_{20} \\ C_{21} &= \text{cum}(y(k), y^*(k)) = M_{21} \\ C_{40} &= \text{cum}(y(k), y(k), y(k), y(k)) \\ &= M_{40} - 3M_{20}^2 \\ C_{41} &= \text{cum}(y(k), y(k), y(k), y^*(k)) \\ &= M_{41} - 3M_{20}M_{21} \\ C_{42} &= \text{cum}(y(k), y(k), y^*(k), y^*(k)) \\ &= M_{42} - M_{20}^2 - 2M_{21}^2 \\ C_{60} &= M_{60} - 15M_{40}M_{20} + 30M_{20}^3 \end{aligned}$$

$$C_{63} = M_{63} + 18M_{20}^2M_{21} - 6M_{20}M_{40} - 9M_{42}M_{21} + 12M_{21}^3 \quad (3)$$

where $M_{pq} = E\{y(k)^{p-q}y^*(k)^q\}$ is the moment of $y(k)$. In practical applications, M_{pq} is usually estimated with N samples of $y(k)$

$$M_{pq} = \frac{1}{N} \sum_{k=1}^N y(k)^{p-q} y^*(k)^q. \quad (4)$$

Other HOCs, such as C_{80} , C_{81} , C_{82} , C_{83} , C_{84} , $C_{10,0}$, $C_{10,1}$, and $C_{10,2}$, can be similarly obtained with estimated moments as in [27] and [31].

In [32], an example that utilizes three features of the fourth-order cumulants (C_{40} , C_{41} , and C_{42}) to classify five modulation schemes, including binary phase shift keying (BPSK), 8PSK, four-quadrature amplitude modulation (4QAM), 16QAM, and 64QAM, is illustrated. A three-layer fully connected DNN based on autoencoders is exploited as the classifier. Pretrain of the network is conducted step by step using a greedy layerwise approach. The last autoencoder layer is connected to a softmax layer, which is trained in a supervised manner and able to determine the modulation scheme of the received signal.

In [33], six HOCs (C_{20} , C_{21} , C_{40} , C_{41} , C_{42} , and C_{60}) are extracted as the feature input of DNN to achieve modulation classification of two-frequency shift keying (FSK), 4FSK, BPSK, QPSK, two-amplitude shift keying (ASK), and 4ASK. A four-layer DNN with two hidden layers is leveraged as the modulation classifier. Simulation results demonstrate that HOCs can achieve good classification accuracy even at low signal-to-noise ratio (SNR) scenarios.

The impacts of different cumulants are analyzed in [34], where the meaningless features are abandoned and only the effective features are adopted to reduce the calculation time for potential real-time applications. For instance, C_{40} is the most effective and C_{20} is the most meaningless among the cumulants set $\{C_{20}, C_{21}, C_{40}, C_{41}, C_{42}\}$ to classify BPSK, QPSK, 8PSK, 16QAM, and 64QAM. A DNN consisting of an input layer, two hidden layers, and an output layer is adopted to learn from these cumulants and up to 8% of classification accuracy improvement is achieved by using the effective cumulants.

B. Spectral Features

Spectral features, such as amplitude features and phase features, are another group of characteristics that can be adopted for modulation classification [35], [36]. To calculate the spectral features, the received signal $y(k)$ is rewritten as follows:

$$\begin{aligned} y(k) &= I(k) + j \cdot Q(k) \\ &= a(k) \cdot e^{j\phi(k)} \end{aligned} \quad (5)$$

where $I(k)$, $Q(k)$, $a(k)$, and $\phi(k)$ represent the instantaneous in-phase component, quadrature component, amplitude, and phase of $y(k)$, respectively.

The ratio of powers of quadrature component and in-phase component gives [37]

$$\beta = \frac{\sum_{k=1}^N Q^2(k)}{\sum_{k=1}^N I^2(k)}. \quad (6)$$

The average value of the instantaneous amplitude is expressed as

$$m_a = \frac{1}{N} \sum_{k=1}^N a(k) \quad (7)$$

and the normalized and normalized-centered instantaneous amplitudes are given by

$$\begin{aligned} a_n(k) &= a(k)/m_a \\ a_{nc}(k) &= a_n(k) - 1. \end{aligned} \quad (8)$$

The peak-to-average ratio (PAR) of the amplitude is

$$R_{PA} = \max\{a(k)\}/m_a \quad (9)$$

and the peak-to-root-mean-square ratio (PRR) of the amplitude is

$$R_{PR} = \max\{a^2(k)\} / \frac{1}{N} \sum_{k=1}^N a^2(k) \quad (10)$$

where $\max\{\cdot\}$ returns the maximum value.

The maximum value of the power spectral density of the normalized-centered instantaneous amplitude is defined as [35], [38]

$$\gamma_{\max} = \max\{|DFT(a_{nc}(k))|^2\}/N \quad (11)$$

where $DFT(\cdot)$ is the discrete Fourier transform.

The standard deviation of the absolute value of the normalized instantaneous amplitude is

$$\sigma_{aa} = \sqrt{\frac{1}{N} \left(\sum_{k=1}^N a_{nc}^2(k) \right) - \left(\frac{1}{N} \sum_{k=1}^N |a_{nc}(k)| \right)^2}. \quad (12)$$

Similarly, the standard deviation of the absolute value of the nonlinear component of the instantaneous phase is

$$\sigma_{ap} = \sqrt{\frac{1}{C} \left(\sum_{a_n(k) > a_t} \phi_{NL}^2(k) \right) - \left(\frac{1}{C} \sum_{a_n(k) > a_t} |\phi_{NL}(k)| \right)^2} \quad (13)$$

where $\phi_{NL}(k)$ is the value of nonlinear component of the instantaneous phase, C is the number of samples in set $\{y(k)\}$ for which $a_n(k) > a_t$, and a_t is a threshold for $a(k)$ below which the estimation of the instantaneous phase is very sensitive to the noise.

The standard deviation of the nonlinear component of the direct (not absolute) instantaneous phase is equal to

$$\sigma_{dp} = \sqrt{\frac{1}{C} \left(\sum_{a_n(i) > a_t} \phi_{NL}^2(i) \right) - \left(\frac{1}{C} \sum_{a_n(i) > a_t} \phi_{NL}(i) \right)^2}. \quad (14)$$

The skewness and kurtosis of the normalized instantaneous amplitude are evaluated by [39], [40]

$$Sk = \frac{1}{N} \sum_{k=1}^N a_n^3(k) / \left(\frac{1}{N} \sum_{k=1}^N a_n^2(k) \right)^{3/2} \quad (15)$$

$$Ku = \frac{1}{N} \sum_{k=1}^N a_n^4(k) / \left(\frac{1}{N} \sum_{k=1}^N a_n^2(k) \right)^2. \quad (16)$$

In [31], as many as 28 features, including both cumulants and spectral features, are combined to train a DNN with four fully connected layers. Since high-dimensional feature space is considered, the proposed algorithm is able to achieve excellent classification accuracy of BPSK, QPSK, 8PSK, 16QAM, and 64QAM in fading environments.

In [37], total 12 cumulants and spectral features are extracted from the received signal and fed into the DNN for modulation classification. Normalization, which maps data between 0 and 1, is applied on each feature to make the weight of feature dimension consistent with the objective function and accelerate the convergence of the iterative solution. Particle swarm optimization is utilized to optimize the number of hidden layer nodes of the DNN to avoid local minimization and manual node selection.

C. Circular Features

The features based on circular summary statistics are useful to separate FSK-type and QAM-type signals [41]. Specifically, the q th-order cyclic moment of the instantaneous phase is defined as

$$v_q = \frac{1}{N} \sum_{k=1}^N e^{jq\phi(k)}. \quad (17)$$

Then, the circular summary statistics-based features named circular variance, circular skewness, and circular kurtosis are given by

$$\zeta_{\text{cir}} = 1 - |v_1| \quad (18)$$

$$\eta_{\text{cir}} = \frac{|v_2| \sin(\angle v_2 - 2\angle v_1)}{(\zeta_{\text{cir}}^2)^{\frac{3}{2}}} \quad (19)$$

$$\zeta_{\text{cir}} = \frac{|v_2| \cos(\angle v_2 - 2\angle v_1) - |v_1|^4}{(\zeta_{\text{cir}}^2)^2} \quad (20)$$

where $\angle v_q$ is the direction of v_q .

In [36], a hybrid feature space comprising cumulants (C_{40} , C_{41} , and C_{42}), spectral features (β , σ_{ap} , and σ_{dp}), and circular features (ζ_{cir} , η_{cir} , and ζ_{cir}) is suggested on the basis of empirical evaluation of their statistical behavior to distinguish constant envelope modulations, e.g., Gaussian filtered minimum shift keying (GMSK), Gaussian FSK (GFSK), continuous-phase FSK (CPFSK), and offset QPSK (OQPSK). Radial basis function network (RBFN) and sparse autoencoder-based DNN are proposed as the classifiers. Both networks perform well with the hybrid features.

Moreover, besides the cumulants, spectral, and circular features listed above, other types of features, such as cyclostationary analysis-based features [21], [35], wavelet transform-based features [35], [42], and approximate entropy-based features [22], can be fed into DNNs as well to achieve DL-based modulation classification with feature representation.

Among all the DL-based algorithms with feature representation, one remarkable example is given in [31], which exploits as many as 28 features derived from 20000 samples (2000 symbols) to classify five modulation schemes and achieves an average accuracy of 86% at 5 dB SNR in a Rayleigh fading channel with 100 Hz Doppler frequency

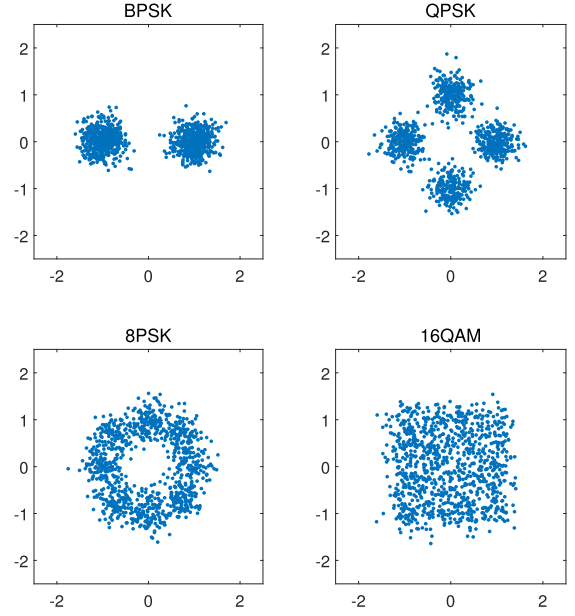


Fig. 2. Constellation images of four modulation schemes at SNR = 10 dB.

offset. According to this example, more features tend to lead to better performance.

IV. DL METHODS: IMAGE REPRESENTATION

DL is a powerful classification tool originated in the field of computer vision. From the perspective of computer vision, one direct approach of exploiting DL for modulation classification is to convert the received signal into an image and utilize DNNs to recognize the image, which leads to DL-based modulation classification with image representation.

A. Constellation Diagram

Constellation diagram is a widely used 2-D image that maps signal samples into scattering points on a complex plane, as shown in Fig. 2. The feasibility of exploiting constellation diagram and DNNs for modulation classification is demonstrated in [9], which converts 1000 samples of the received signal into a 227×227 binary constellation image and recognizes the image using a convolutional neural network (CNN) AlexNet [43] to classify QPSK, 8PSK, 16QAM, and 64QAM. In [44], a graphic constellation projection algorithm with low computation complexity is proposed to convert the received signal into a 25×25 constellation image. A deep belief network (DBN) [45] is adopted to learn the underlying features in these constellation diagrams and recognize the candidate modulation schemes of BPSK, QPSK, 8PSK, and 16QAM.

In order to alleviate the information loss resulted from signal conversion, enhanced gray and three-channel images of constellation diagram are suggested in [10]. The enhanced gray constellation image takes both the location variations of different samples within one pixel and the impacts of each sample on its neighboring pixels into consideration with an exponential decay model

$$B_{i,j} = P \cdot e^{-\lambda \cdot d_{i,j}} \quad (21)$$

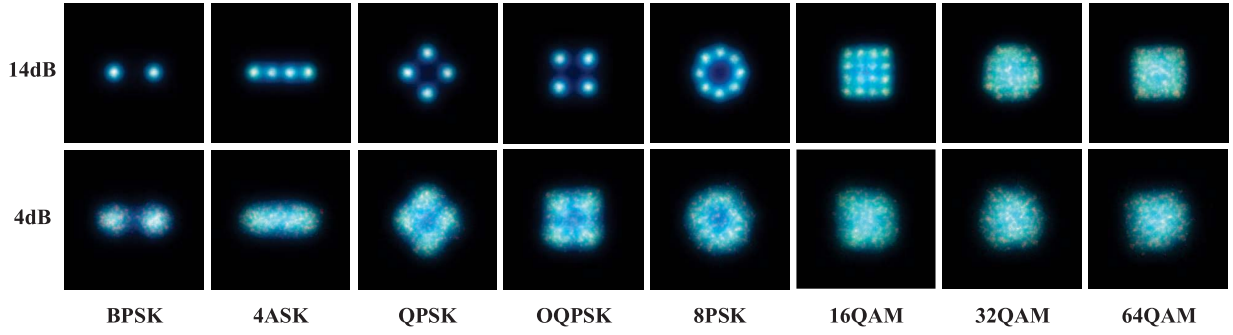


Fig. 3. Three-channel images of eight modulation schemes at different SNRs [10].

where $B_{i,j}$ denotes the impact of a sample point $\#i$ on Pixel $\#j$, P is the power of each sample point, $d_{i,j}$ is the distance between sample point $\#i$ and the centroid of Pixel $\#j$, and λ is an exponential decay rate. The three-channel image comprises three enhanced gray images derived from the same batch of signal samples but different exponential decay rates, as shown in Fig. 3. The constellation images are fed into two CNNs of AlexNet and GoogLeNet [46] for modulation classification. Higher classification accuracy is achieved compared with the binary and gray constellation diagrams. Furthermore, the appropriate size of the selected area of complex plane to generate the constellation image and the impact of the image resolution is also investigated in [10].

Another type of constellation diagram, named contour stellar image, is proposed to take full advantage of the pixel density and color [47]. This image uses a square window to slide on constellation diagram and counts how many sample points are located in its field. Its relative point density is calculated as

$$\rho(i, j) = \frac{\sum_{i=x_1}^{x_2} \sum_{j=y_1}^{y_2} p(i, j)}{\sum_{x_1=W_0}^{W_1} \sum_{y_1=H_0}^{H_1} \sum_{i=x_1}^{x_2} \sum_{j=y_1}^{y_2} p(i, j)} \quad (22)$$

where (x_1, y_1) and (x_2, y_2) stand for the coordinates of top-left and bottom-right corners of the sliding window, respectively, (W_0, H_0) and (W_1, H_1) stand for the coordinates of top-left and bottom-right corners of the constellation diagram, respectively, and $p(i, j)$ indicates a sample point with coordinate (i, j) in constellation diagram. The relative point density $\rho(i, j)$ can be normalized and mapped into a color bar and a contour stellar image is consequently generated. Besides the images obtained from the original received signal, additional contour stellar images can be generated using generative adversarial networks (GANs) to enlarge the training dataset and improve the classification performance.

In [48], the contrast of constellation diagram is enhanced to highlight the regions where the signal samples are notably dense. Since the pixel value of grayscale images is in a range of 0–255, it is easy to enhance the image contrast by grayscale expanding and alternate the pixel value into

$$P_f = \begin{cases} 255p_1, & p < p_1 \\ 255 \left[\frac{q_2 - q_1}{p_2 - p_1} (p - p_1) + q_1 \right], & p_1 \leq p \leq p_2 \\ 255p_2, & p > p_2 \end{cases} \quad (23)$$

where q is the ratio of the sample number in the pixel to the total sample number, $p_1 = 0.3$ and $p_2 = 0.7$ are the original intervals, and $q_1 = 0.1$ and $q_2 = 1$ are the intervals of enhanced images. Both the regular and contrast-enhanced constellation images are input into a compressive CNN that jointly exploits softmax loss and compressive loss for modulation classification. Simulation results show that 0.5 dB SNR gain can be obtained at the target accuracy of 90% compared with the algorithm that only utilizes regular constellation images.

Based on the constellation diagrams discussed above, more techniques have been applied to handle the task of DL-based modulation classification. In [49], a contrastive fully convolutional network is designed to learn high-dimensional representations of each modulation scheme and provide robust classification performance. In [50], the second- and fourth-order moment (M_2M_4)-based estimator and the multilabel DL are adopted to address the issue of varying SNR. In order to deploy the DL-based modulation classification in edge devices and real-time scenarios, neural network pruning is exploited to reduce the convolution parameters and floating-point operations with little accuracy loss [51]. In [52], masking filters are implemented to discard constellation points corrupted due to significant noise.

B. Eye Diagram

Eye diagram is an oscilloscope display where the received signal is repetitively applied to the vertical input and the data rate is used to trigger the horizontal sweep. In [53], colored eye diagrams are collected from the oscilloscope with resolution 900×1200 , as shown in Fig. 4. To reduce the computation amount and enhance the generalization ability, colored images are transformed into grayscale and down-sampled to 28×28 . Then, the gray eye diagrams are fed into a CNN, with which four modulation schemes of return-to-zero on-off keying (RZ-OOK), nonreturn-to-zero OOK (NRZ-OOK), RZ-differential phase shift keying (RZ-DPSK), and four-pulse amplitude modulation (4PAM) are classified.

C. Feature Point Image

Feature point (FP) image is generated from signal features, such as HOCs, PAR, PRR, maximum value of power spectral density, and skewness, which can be derived from (3), (9), (10), (11), and (15), respectively. In [40], these features are

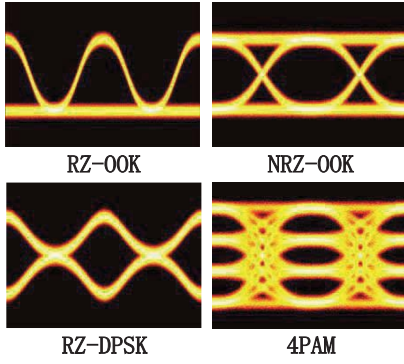


Fig. 4. Eye diagrams of four modulation schemes from oscilloscope [53].

plotted on a complex plane within a square area of $[-3.5, 3.5]$. The complex feature values, including C_{20} , C_{40} , and C_{61} , are directly plotted without additional transformation. The real feature values, including C_{21} , C_{42} , R_{PA} , R_{PR} , $\gamma_{\max}$, and Sk , are transformed into complex FPs as follows:

$$FP_1 = C_{21} + j3C_{42} \quad (24)$$

$$FP_2 = -4Sk + j(R_{PR} - 1) \quad (25)$$

$$FP_3 = -3\gamma_{\max} - jR_{PA}. \quad (26)$$

Fig. 5 shows the FP images of BPSK, QPSK, 8PSK, 4PAM, 16QAM, and 64QAM at SNR of 10 dB. Since different modulation schemes have different FP images, Lee *et al.* [40] utilized a CNN to recognize these FP images and performed modulation classification. Simulation results show that the DL-based modulation classification using FP images has superior performance when SNR is high.

D. Ambiguity Function Image

Ambiguity function (AF), a bivariate function with respect to time delay τ and frequency offset ω , is defined by

$$AF_s(\omega, \tau) = \int_{-\infty}^{\infty} y\left(t + \frac{\tau}{2}\right) y^*\left(t - \frac{\tau}{2}\right) e^{-j\omega t} dt \quad (27)$$

where $y(t)$ is the continuous received signal and $*$ denotes the complex conjugate. Considering the different values of ω and τ , a 2-D matrix of $AF_s(\omega, \tau)$ can be obtained according to (27) and saved as an AF image.

Different modulation schemes, such as ASK, PSK, QAM, FSK, MSK, linear frequency modulation (LFM), and orthogonal frequency-division multiplexing (OFDM), exhibit different AF images, as shown in Fig. 6. In [54], 28×28 AF images with normalized values are cut from the central part of the whole AF images and fed into sparse autoencoders, which determines the modulation scheme of the received signal.

E. Spectral Correlation Function Image

Modulated signals are usually treated as cyclostationary processes with periodic characteristics depicted by cyclic autocorrelation function (CAF) and spectral correlation function (SCF). The CAF, which quantizes the amount of correlation

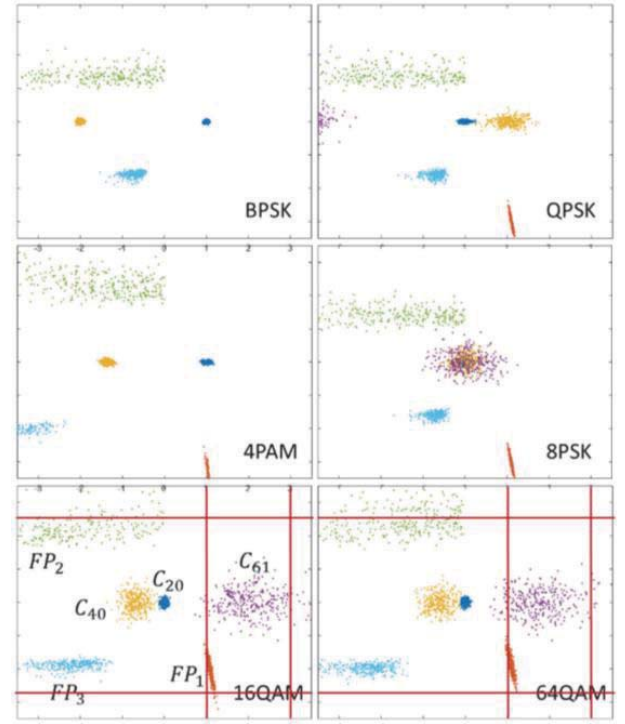


Fig. 5. FP images of six modulation schemes at SNR = 10 dB [40].

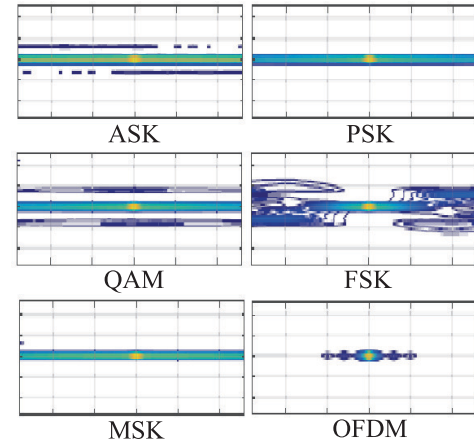


Fig. 6. AF images of different modulation schemes at SNR = 5 dB [54].

between different frequency shifts and represents the fundamental parameters of the second-order periodicity, is estimated by

$$R(l, \alpha) = \left[\lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{k=-N}^N y(k) y^*(k-l) e^{-j2\pi \alpha k} \right] e^{-j\pi \alpha l} \quad (28)$$

where α is the cyclic frequency indicating the cyclic evolution of the waveforms. The SCF is derived from CAF via Fourier transform

$$S(f, \alpha) = \sum_{l=-\infty}^{\infty} R(l, \alpha) e^{-j2\pi f l}. \quad (29)$$

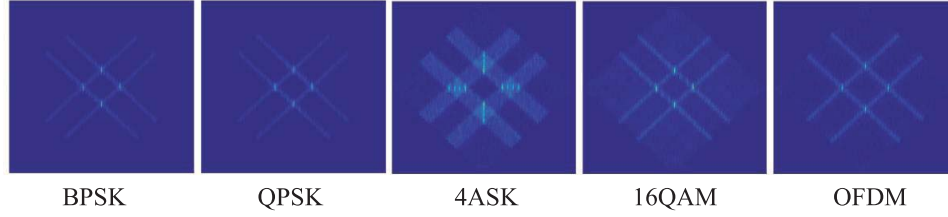


Fig. 7. SCF images of five modulation schemes [55].

Note that SCF is a bivariate function with respect to digital frequency f and cyclic frequency α . Considering the discrete values of f and α ranging in $[-0.5, 0.5]$ and $[-1, 1]$, respectively, an SCF matrix can be derived according to (29). Taking the absolute value of SCF matrix, a real matrix is obtained and regarded as the SCF image, as shown in Fig. 7 [55]. Then, the SCF image is scaled into 64×64 pixels using the 2-D fast Fourier transform (FFT) and inverse FFT. Due to the symmetric and sparse property of SCF image, a triangle part with $1/8$ pixels of the scaled SCF image is selected to reduce the calculation complexity. After that, the selected 64×64.8 pixels are reshaped into a vector with the length of 512. Finally, the vector is normalized to zero mean and unit variance and serves as the input of DBN. In [56], the activation function of DBN is approximated to realize the digital logic implementation of the proposed algorithm without multiplication operations.

F. Cyclic Correntropy Spectrum Graph

The correntropy function of the received signal $y(t)$ is defined as

$$V_x(t, \tau) = E\{\kappa_\sigma(y(t) - y(t + \tau))\} \quad (30)$$

where σ represents the kernel size and the Gaussian kernel $\kappa_\sigma(y(t) - y(t + \tau))$ is expressed as

$$\kappa_\sigma(y(t) - y(t + \tau)) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(y(t) - y(t + \tau))^2 / 2\sigma^2}. \quad (31)$$

Then, the cyclic correntropy (CCE) of the received signal is given by

$$C_x(\xi, \tau) = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} V_x(t, \tau) e^{-j2\pi\xi t} dt \quad (32)$$

where T_0 is the period of $V_x(t, \tau)$ and ξ is the cyclic frequency. Accordingly, the cyclic correntropy spectrum (CCES) is obtained via the Fourier transform of CCE

$$S_x(\xi, f) = \int_{-\infty}^{+\infty} C_x(\xi, \tau) e^{-j2\pi f \tau} d\tau. \quad (33)$$

Since CCES is mostly not affected by the impulsive noises and exhibits significant differences with respect to different modulation schemes, Ma *et al.* [57] suggested its use for modulation classification under burst non-Gaussian noise environments. In order to make the CCES features more prominent, logarithmic transformation is applied on CCES

$$LS_x(\xi, f) = \log(S_x(\xi, f) + 10^{-6}). \quad (34)$$

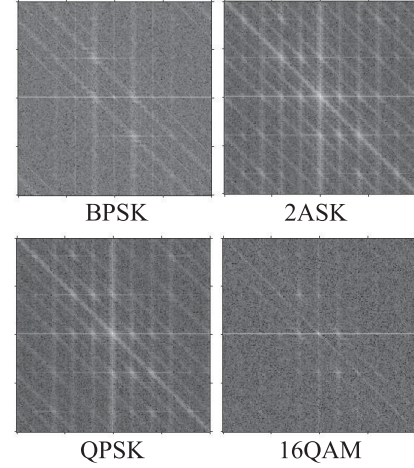


Fig. 8. Log-CCES graphs of four modulation schemes [57].

The log-CCES graphs of BPSK, 2ASK, QPSK, and 16QAM modulated signals are shown in Fig. 8. ResNet [58], a famous CNN in computer vision, is used to recognize these graphs. Simulation results show that the proposed algorithm is able to combat the non-Gaussian noises and achieves high classification accuracy as expected.

G. Bispectrum Graph

Bispectrum is the 2-D sequence Fourier transform of the third-order cumulant when the sum of the third-order cumulant is finite. The amplitude spectrum of bispectrum (ASB) is given by

$$|B_y(\omega_1, \omega_2)| = |X(\omega_1)| \cdot |X(\omega_2)| \cdot |X(\omega_1 + \omega_2)| \quad (35)$$

where ω_1 and ω_2 represent two frequency indexes and $X(\omega) = \sum_{k=-\infty}^{+\infty} y(k) e^{-j\omega k}$.

Fig. 9 shows the ASBs of seven modulation schemes, including BPSK, 2ASK, 2FSK, 4FSK, 8FSK, LFM, and OFDM, with respect to ω_1 and ω_2 . In [59], these graphs are resized to 227×227 and recognized by AlexNet. The output of AlexNet accurately indicates the modulation scheme of the received signal.

With the help of image representation, the received signals are converted into various images. Note that these images can be further processed before being fed into DNNs. In [10], the original constellation diagram is processed into a three-channel image and the classification accuracy of 8PSK is improved from 82.8% to 97.1% at 2 dB SNR over the

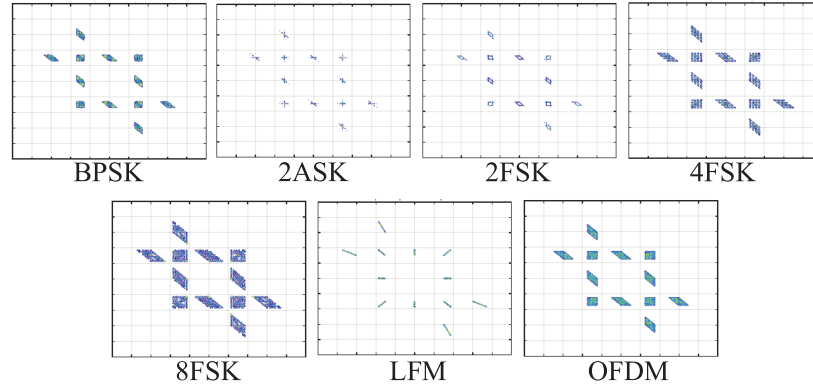


Fig. 9. ASB graphs of seven modulation schemes [59].

additive white Gaussian noise (AWGN) channel. In [48], the use of contrast-enhanced grid constellation diagram to classify five modulation schemes obtains 0.5 dB SNR gain at the target accuracy of 90% over the Rayleigh fading channel. Therefore, image processing is often beneficial for the image representation method.

V. DL METHODS: SEQUENCE REPRESENTATION

In communications systems, signal samples are sequentially received. This section discusses the DL-based modulation classification with sequence representation.

A. In-Phase and Quadrature Sequences

The exploitation of in-phase and quadrature (IQ) sequences for DL-based modulation classification is first proposed in [7], where the received signals of eight digital and three analog modulation schemes are identified [60]. The signals are segmented through a 128-point rectangular window. Each segment consists of a 128-point in-phase sequence and a 128-point quadrature sequence. Fig. 10 shows the IQ sequences of eight digital modulation schemes [61]. These sequences are fed into two CNNs and one fully connected DNN separately. Both CNNs comprise two convolutional layers as well as two dense layers, and the DNN comprises four dense layers. In low SNR scenarios with a short sequence length, all networks demonstrate powerful ability of modulation classification.

In [8], the dataset of IQ sequences is extended with an improved version of software-defined radio. More modulators are considered to cover a wider range of single-carrier modulation schemes. The new datasets have two different compositions named “Normal” and “Difficult,” which involve 11 and 24 modulation schemes, respectively. Several propagation scenarios of simulated wireless channels and over-the-air transmission channels are also discussed. Moreover, the length of IQ sequences increases from 128 to 1024. Two CNNs, VGGNet [62] and ResNet, are implemented to learn from these sequences, and effective modulation classification can be achieved if sufficient training data are available.

Based on the above datasets, more techniques and networks have been developed for DL-based modulation classification. In [63], three methods of data augmentation are analyzed to

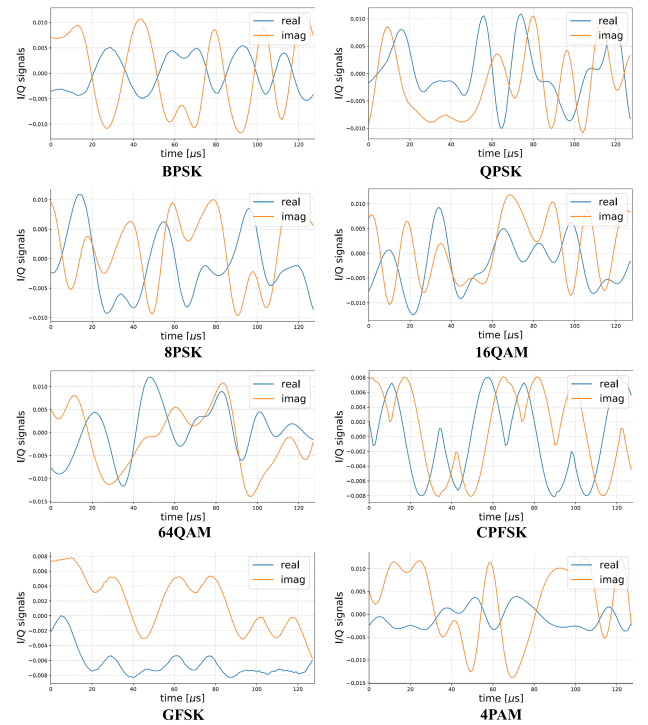


Fig. 10. IQ sequences of eight digital modulation schemes [61].

enlarge the datasets and avoid the overfitting problem. The feasibility of four networks, including a seven-layer CNN, a seven-layer ResNet, a seven-layer DenseNet [64], and an eight-layer convolutional long short-term DNN (CL-DNN) [65], is demonstrated in [66]. Comparisons between CNN and long short-term memory (LSTM) network [67] are conducted in [68], where the superiority of the latter in classification accuracy is observed. Considering the future applications with time and resource constraints, Ma *et al.* [69] constructed a hybrid network composed of CNN and LSTM, based on which a lightweight network with shallow layers and few nodes is trained to reduce the inferring time and storage consumption. In [70], a neural network-based signal distortion correction module (CM) is added to eliminate the frequency and phase offset introduced by the wireless channel. In [71], an improved

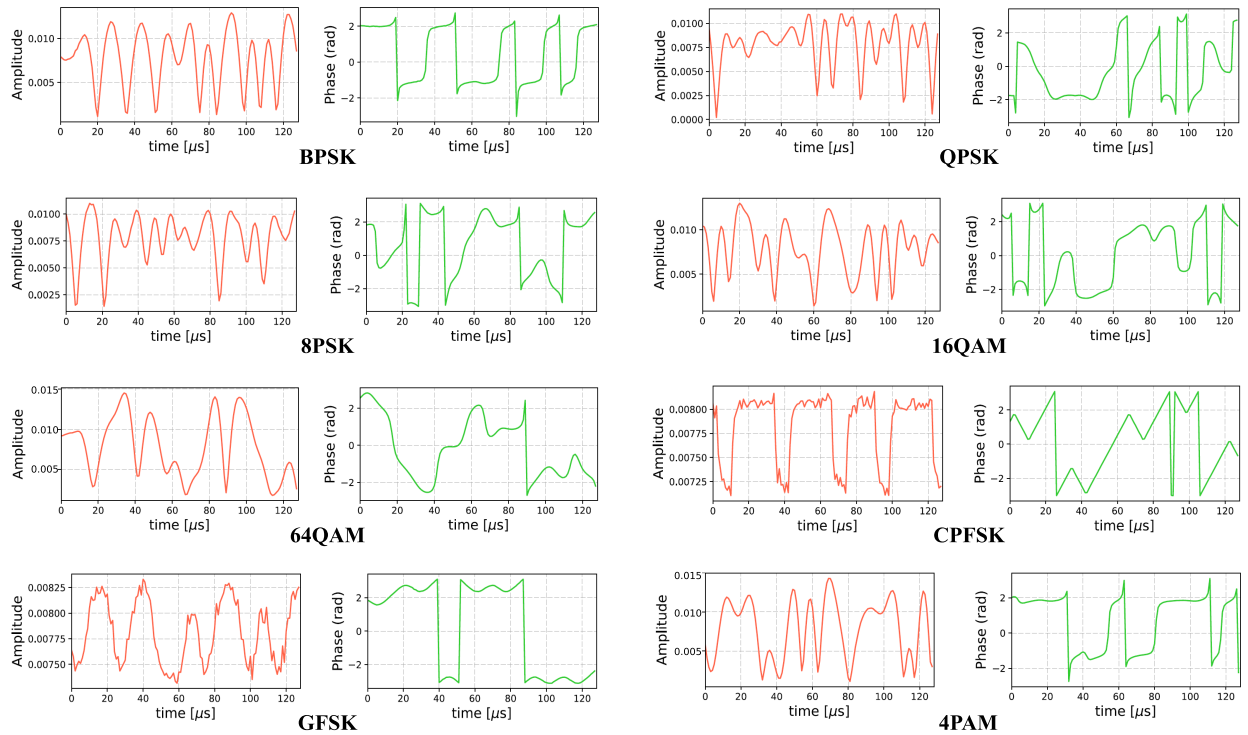


Fig. 11. AP sequences of eight digital modulation schemes [61].

CNN with different layers and fewer filters is proposed to comply with the estimated beyond-5G requirements on latency.

Similar examples of DL-based modulation classification using other datasets of IQ sequences can be found in [72]–[79]. According to [72], IQ sequences of 4PSK, 16PSK, 16QAM, and 128QAM modulated signals are generated using MATLAB and input into an unsorted DNN. In [73], IQ sequences in OFDM systems are handled by a CNN consisting of two convolutional layers and three fully connected layers to achieve accurate, consistent, and robust modulation classification. Moreover, in [74], the harmful effect of carrier phase offset in OFDM systems is considered and eliminated to ensure high classification accuracy. In [75], overlapped IQ sequences are divided into multiple subsequences to accommodate the varying sequence length and transfer learning is applied to improve retraining efficiency. Three fusion rules, including voting-based fusion, confidence-based fusion, and FB fusion, are utilized to combine the results of multiple subsequences and obtain better classification accuracy [76]. In order to apply modulation classification in general scenarios with multiple-channel types, an extra CNN is inserted before the modulation classifier to recognize the channel type [77]. In [78], an RNN-based classifier is introduced to combat the Rayleigh fading as well as uncertain noise and achieve robustness to the increase of frequency offset. Cooperative modulation classification using multiple IQ sequences received by multiple antennas is studied in [79].

B. Amplitude and Phase Sequences

Besides IQ-sequence representation, the received signal can also be characterized by amplitude and phase (AP). In [61],

each segment of a signal [60] is mapped into a 128-point amplitude sequence and a 128-point phase sequence, as shown in Fig. 11. The network adopted to process these sequences is similar to that of [7] for a fair comparison. Experimental results reveal that the AP-sequence representation leads to performance improvements up to 2% for medium to high SNRs compared with the IQ-sequence representation. Moreover, in [80], LSTM networks are exploited to learn from the datasets mentioned in [8] and [60] represented in AP sequences, which achieves high classification accuracy and is able to tackle the issue of variable sample rate and sequence length.

C. Fast Fourier Transformation Sequences

The FFT representation is a mapping from the time-domain IQ signal into frequency-domain complex data consisting of two real-valued sequences, which carry the real and imaginary parts of FFT. In [61], 128-point FFT is performed on the dataset mentioned in [60]. The magnitudes of FFT sequences of eight digital modulation schemes are shown in Fig. 12. These sequences are fed into a CNN, with which the feasibility of modulation classification with FFT-sequence representation is confirmed.

In order to highlight the superiority of FFT-sequence representation, Mossad *et al.* [81] compared it with the IQ-sequence representation. Experimental results indicate that, for high-order modulation schemes, such as QPSK, 8PSK, 16QAM, and 64QAM, FFT-sequence representation produces better classification accuracy than IQ-sequence representation.

Another advantage of FFT-sequence representation is shown in [80]. Note that the FFT sequences could be short if a

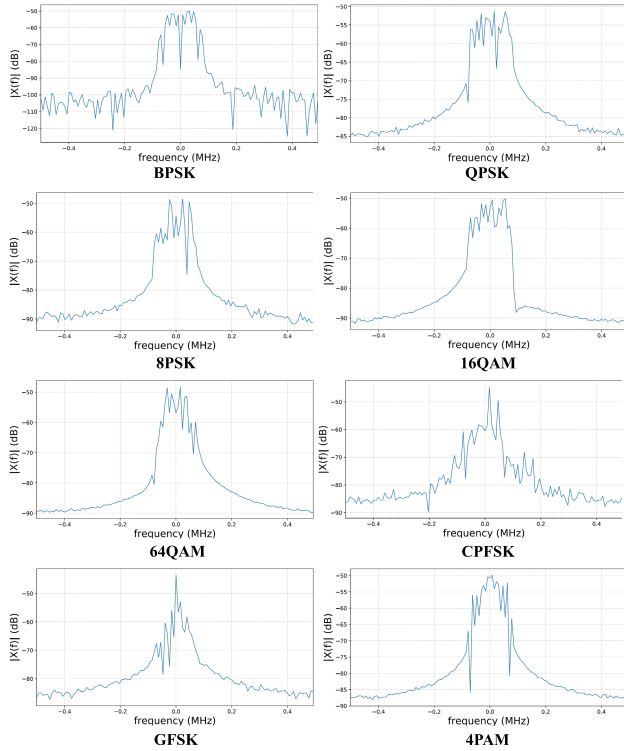


Fig. 12. Magnitude of FFT sequences of eight digital modulation schemes [61].

small size of FFT is used, resulting in low consumption of computation resources. Therefore, modulation classification with FFT-sequence representation could be suitable for low-complexity applications, e.g., electrosense network [82], where the distributed sensing nodes are not powerful enough to process massive IQ data.

D. Amplitude Histogram Sequences

Histogram is an accurate representation constructed by dividing the entire range of data into a series of intervals and counting how many data fall into each interval. Fig. 13 shows the amplitude histograms of QPSK, 16QAM, and 64QAM modulated signals at different SNRs [83], [84]. As shown in Fig. 13, the shapes of amplitude histograms are unique and contain distinctive signatures of modulation schemes, which may be recognized by DNNs to achieve modulation classification.

In [83], 56000 samples of the received signal after constant-modulus-algorithm equalization are collected to synthesize each amplitude histogram with 80 intervals. Indeed, each amplitude is an 80-point sequence. A large number of histogram sequences are obtained and fed into a DNN with two hidden layers for training. The trained network is able to identify QPSK, 16QAM, and 64QAM accurately.

Some interesting results of sequence representation can be found in [61], where IQ sequence, amplitude/phase sequence, and FFT sequence are tested and compared based on the same CNN and the same dataset of RadioML 2016.10a [7]. When the SNR is high enough, amplitude/phase sequence achieves the best accuracy of 84%, better than IQ sequence (81%) and

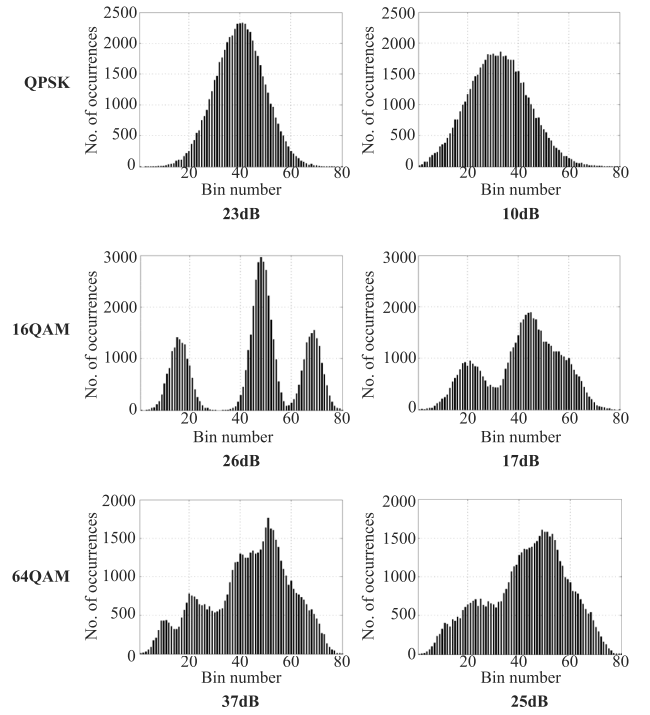


Fig. 13. Amplitude histograms of three modulation schemes at different SNRs [83].

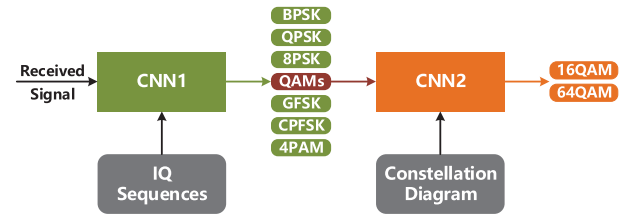


Fig. 14. Architecture of modulation classification using IQ sequences and constellation diagram [85].

FFT sequence (69%). However, in the medium SNR region of 2–5 dB, the IQ sequence performs the best. If the SNR is very low, three sequences have similar performance. Consequently, it is important to choose an appropriate sequence representation according to the environment.

VI. DL METHODS: COMBINED REPRESENTATION

As stated in Sections III–V, a received signal can be represented by either features, images, or sequences. This section considers the DL-based modulation classification that utilizes combined representation as the input of DNN.

A. Features and Sequences

Inspired by the DL-based color image classification that uses three-channel information, the handcrafted features of fourth-order cumulants C_{40} , C_{41} , and C_{42} are added as the third component of signal representation combined with IQ sequences [68]. Therefore, the received signal is represented in a format of 3×128 , which comprises a 128-point in-phase sequence, a 128-point quadrature sequence, and a 128-point

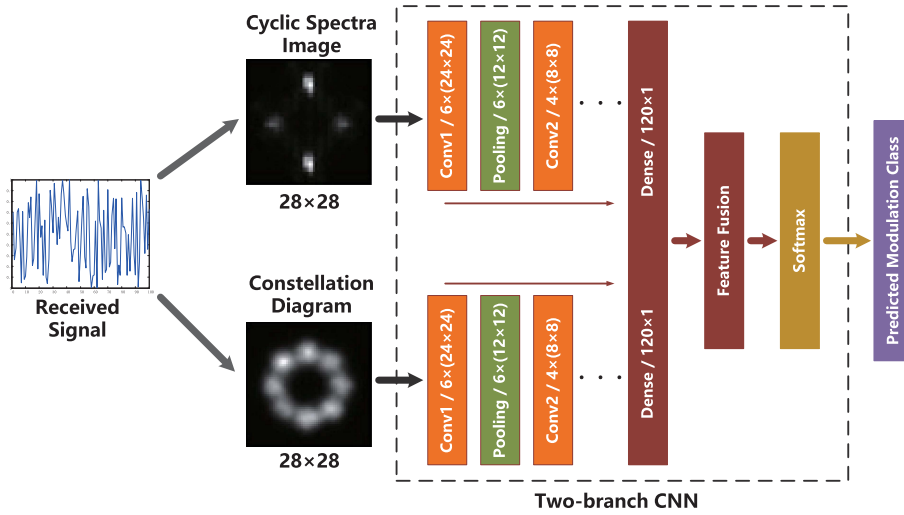


Fig. 15. Architecture of modulation classification using cyclic spectra image and constellation diagram [86].

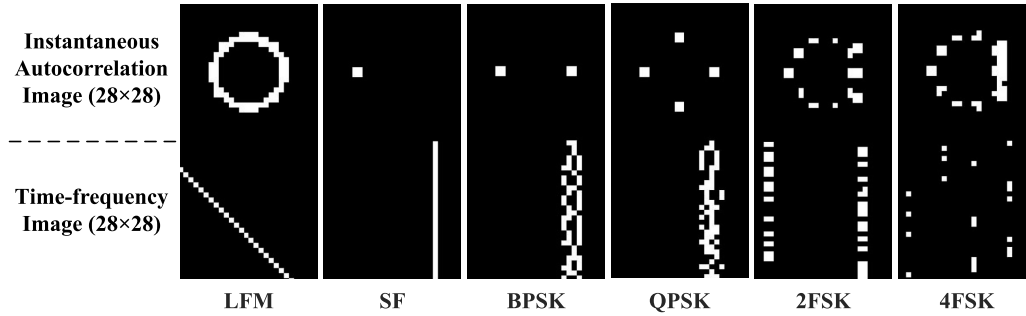


Fig. 16. Joint instantaneous autocorrelation and time-frequency images of six modulation schemes [87].

cumulants feature. Both CNN and LSTM network are utilized to learn from this combined representation, which results in roughly 8% of accuracy improvement in modulation classification and better robustness to noises.

B. Image and Sequences

Since DL-based modulation classification with IQ-sequence representation is often confused when classifying 16QAM and 64QAM in low SNR environments, Wang *et al.* [85] proposed a new algorithm utilizing both IQ sequences and constellation diagram to achieve accurate classification of eight modulation schemes, including BPSK, QPSK, 8PSK, GFSK, CPFSK, 4PAM, 16QAM, and 64QAM. The architecture of the proposed algorithm is shown in Fig. 14.

According to Fig. 14, two CNNs are adopted for modulation classification. The first CNN consisting of two convolution layers and four fully connected layers is trained with IQ sequences to distinguish the modulation schemes of BPSK, QPSK, 8PSK, GFSK, CPFSK, and 4PAM. The second CNN containing more trainable parameters and an average pooling operation is trained with constellation diagram to identify 16QAM and 64QAM, which are hardly recognized by the first CNN. Experiment results show that, with the use of the combined representation of IQ sequences and constellation

diagram, all eight modulation schemes can be classified with high accuracy.

C. Multiple Images

The cyclic spectra image has been regarded as a good representation of the received signal due to its strong resistibility to channel noises. However, modulation classification using cyclic spectra images usually suffers from performance degradation when the modulation order is high. To solve this problem, a constellation diagram that is superior in classifying high-order modulation is also taken into consideration [86]. As shown in Fig. 15, the received signal is first transformed into two 28×28 images, including a cyclic spectra image and a constellation diagram. Then, a two-branch CNN is built to extract high-level features of both images separately. After that, multifeature fusion is exploited to integrate the extracted features. Finally, softmax regression is used to infer the classification result. Compared with the state-of-the-art algorithms, combined representation of cyclic spectra image and constellation diagram is able to obtain better or a least equivalent classification accuracy with lower computation complexity.

As the time-frequency image is insensitive to phase-modulated signals while the instantaneous autocorrelation

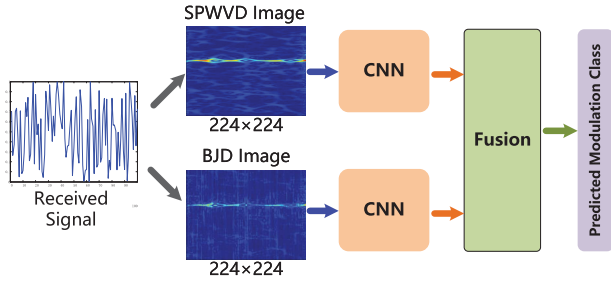


Fig. 17. Architecture of modulation classification using SPWVD and BJD images [88].

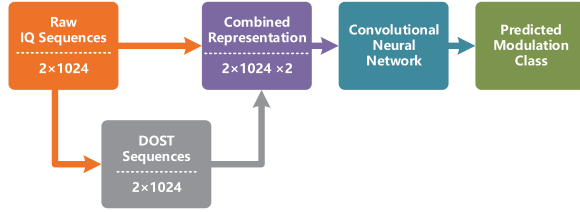


Fig. 18. Architecture of modulation classification using IQ and DOST sequences [90].

image is unclear on frequency-coded signals, a joint image combining these two images is proposed to solve the problem of unified modulation classification [87]. Specifically, the received signal is converted into a time–frequency image and an instantaneous autocorrelation image. Both images are converted to binary images and resized into 28×28 pixels to construct a joint image, whose upper half part and lower half part are the instantaneous autocorrelation image and time–frequency image, respectively, as shown in Fig. 16. The joint image is handled by a seven-layer CNN for modulation classification. Simulation results show that the joint image is significantly superior over each single image in classification accuracy.

Similarly, in [88], the received signal is represented by two 224×224 images via smooth pseudo-Wigner–Ville distribution (SPWVD) and Born–Jordan distribution (BJD), respectively. As shown in Fig. 17, both images are fed into fine-tuned CNNs for feature extraction. Multimodality fusion is exploited to fuse the features of SPWVD and BJD images and achieve accurate modulation classification. Moreover, according to [88], handcrafted features, such as $\gamma_{\max}$, δ_{aa} , δ_{ap} , δ_{dp} , and cumulants, can be fused simultaneously, which leads to further improvement of classification accuracy.

D. Multiple Sequences

Discrete orthonormal Stockwell transform (DOST) [89] is a low-complexity time–frequency decomposition that provides absolutely referenced phase information. In [90], IQ sequences and DOST of IQ sequences are combined to represent the received signal for CNN-based modulation classification, as shown in Fig. 18.

According to Fig. 18, the raw 1024-point IQ sequences of the received signal, derived from the dataset of [8], are

regarded as the first component of signal representation. In addition, IQ sequences are preprocessed via DOST to construct the second component of signal representation, which comprises two 1024-point sequences indicating the absolute values of DOST of in-phase sequence and quadrature sequence. Consequently, the combined representation has four 1024-point sequences and is reshaped into a $2 \times 1024 \times 2$ matrix. The matrix is input into a CNN consisting of five convolutional layers and three fully connected dense layers for modulation classification. With the use of the combined representation of IQ and DOST sequences, CNN is able to make full use of both time- and time–frequency-domain information and improve the overall classification accuracy.

A fair comparison of different algorithms with combined representation can be made based on [68] and [86], both of which use the same dataset of RadioML 2016.10a. In [68], the fourth-order cumulants and IQ sequence are combined, and the average classification accuracy reaches 85% at the SNR of 5 dB. In [86], the cyclic spectra image and constellation diagram are combined, and the accuracy exceeds 90% when the SNR is higher than 5 dB.

VII. SUMMARY AND DISCUSSION

In summary, Table I lists the recent work on DL-based modulation classification according to four groups of signal representations, including feature representation, image representation, sequence representation, and combined representation. For each representation, the received signal has been represented in various formats.

Feature representation is developed originally for the FB algorithms of modulation classification. It extracts multiple features to represent the received signal. Note that the number of extracted features is usually smaller than the length of the received signal. In other words, the received signal is compressed via feature representation, which allows the exploitation of simple DNNs with fewer neurons/layers. However, its drawbacks are also obvious, including the following: 1) the calculation of signal features incurs extra computation complexity; 2) proper features need to be selected according to the candidate modulation set, which requires great expertise and experience; and 3) some key information may be lost when converting the raw signal into features and the performance of modulation classification can be impaired.

The idea of image representation is to transform the received signal into images, with which the task of modulation classification is done via image recognition. Existing DNNs and DL frameworks proposed for image recognition can be directly used for the modulation classification with image representation. Therefore, it is not only easy to implement but also benefits most from the evolution of DL. Moreover, since the signal features are automatically extracted by the hidden layers of DNNs, drawbacks of manual feature extraction are avoided. However, some calculation is still necessary to generate the images and complicated DNNs are required to learn the hierarchical features of images.

Sequence representation is an intuitive type of signal representation as the signal is sequentially received. In communications systems, IQ sequences as well as its equivalent

TABLE I
SUMMARY OF DL-BASED MODULATION CLASSIFICATION WITH DIFFERENT REPRESENTATIONS

Signal Representation	Description	Format	Reference
Feature	Total 28 Features: Second-order, Fourth-order, Sixth-order, Eighth-order and Tenth-order Cumulants, Spectral Features and Statistics Features	1×28	Lee et al. [31]
	Fourth-order Cumulants	1×3	Ali and Yangyu [32]
	Second-order, Fourth-order and Sixth-order Cumulants	1×6	Xie et al. [33]
	Second-order and Fourth-order Cumulants	1×5	Lee et al. [34]
	Fourth-order Cumulants, Circular Summary Statistics Features and Spectral Features	1×9	Shah et al. [36]
	Total 12 Features: Second-order, Fourth-order, Sixth-order and Eighth-order Cumulants, Spectral Features and Statistics Features	1×12	Shi et al. [37]
Image	Constellation Diagram	25×25	Peng et al. [9], [10]
		80×80	Wang et al. [44]
		100×100	Tang et al. [47]
		224×224	Huang et al. [48], [49]
		227×227	Xie et al. [50]
			Tu et al. [51]
			Kumar et al. [52]
	Eye Diagram	28×28	Wang et al. [53]
	Feature Point Image	32×32	Lee et al. [40]
	Ambiguity Function Image	28×28	Dai et al. [54]
	Spectral Correlation Function Image	(64×64)/8	Mendis et al. [55]
Sequence	IQ Sequences	2×128 2×500 2×512 2×1000 2×1024	O'Shea et al. [7], [8]
			Kulin et al. [61]
			Huang et al. [63]
			Liu et al. [66]
			Zhang et al. [68]
			Ma et al. [69]
			Yashashwi et al. [70]
			Hermawan et al. [71]
			Ali and Fan [72]
			Hong et al. [73]
			Shi et al. [74]
			Meng et al. [75]
			Zheng et al. [76]
			Gu et al. [77]
			Hu et al. [78]
			Wang et al. [79]
	Amplitude and Phase (AP) Sequences	2×128	Kulin et al. [61]
			Rajendran et al. [80]
	FFT Sequences	2×128	Kulin et al. [61]
			Mossad et al. [81]
	Amplitude Histogram Sequences	80×1	Khan et al. [83]
Combined	Fourth-order Cumulants and IQ Sequences	3×128	Zhang et al. [68]
	Constellation Diagram and IQ Sequences	128×128+2×128	Wang et al. [85]
	Constellation Diagram and Cyclic Spectra Image	28×28×2	Wu et al. [86]
	Joint Time-Frequency and Instantaneous Autocorrelation Image	(28+28)×28	Wang et al. [87]
	SPWVD and BJD Images	224×224×2	Zhang et al. [88]
	IQ Sequences and DOST Sequences	2×1024×2	Hiremath et al. [90]

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT DL-BASED MODULATION CLASSIFICATION ALGORITHMS

Signal Representation	Description	Reference	DL Network	Channel	Candidate Modulation	SNR (dB)	Accuracy	Complexity
Feature	Total 28 Features	Lee et al. [31]	DNN	AWGN & Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM	[5]	70.64% - 99.95%	/
	Fourth-order Cumulants	Ali and Fan [32]	SAE-DNN	AWGN & Fading	BPSK, 8PSK, 4QAM, 16QAM, 64QAM	[0:5:15]	AWGN: 83% - 100% Phase offset: 98% - 100% Frequency offset: 72%-100%	/
	Second-order & Fourth-order & Sixth-order Cumulants	Xie et al. [33]	Four-layer DNN	AWGN & Fading	2FSK, 4FSK, BPSK, QPSK, 2ASK, 4ASK	[-5,-2]	AWGN: 99.5% - 100% Fading: 90.66% - 100% Frequency Offset: 87.53% - 100%	/
	Second-order & Fourth-order Cumulants	Lee et al. [34]	Four-layer DNN	AWGN	BPSK, QPSK, 8PSK, 16QAM, 64QAM	[-5:5:10]	85.61% - 99.69%	/
	Fourth-order Cumulants & Circular Summary Statics Features & Spectral Features	Shah et al. [36]	RBFN & SAE-DNN	AWGN & Fading	GMSK, CPFSK, GFSK, OQPSK	[0:5:15]	SAE-DNN: AWGN: 71% - 100% Fading: 55% - 96% RBFN: AWGN: 66% - 99% Fading: 50% - 95%	Training Time: SAE-DNN: 121.47s RBFN: 17.22s
	Total 12 Features	Shi et al. [37]	PSO-DNN	AWGN	BPSK, QPSK, 8PSK, 16QAM, 64QAM, 256QAM	[0:2:12]	87% - 100%	/
Image	Enhanced 3-Channel Constellation Diagram	Peng et al. [9], [10]	AlexNet & GoogLeNet	AWGN	BPSK, QPSK, OQPSK, 8PSK, 16QAM, 32QAM, 64QAM, 4ASK	[0:1:10]	74.1% - 100%	(NVIDIA Tesla K40m) Inferring Time: AlexNet: 6.6ms GoogLeNet: 18ms
	Feature Point Image	Lee et al. [40]	Six-layer CNN	AWGN	BPSK, QPSK, 8PSK, 4PAM, 16QAM, 64QAM	[-10:5:10]	17% - 100%	/
	Graphic Constellation Diagram	Wang et al. [44]	DBN	AWGN	BPSK, QPSK, 8PSK, 16QAM	[0:2:10]	95.14% - 100%	Complexity: O(N) (N is the number of symbols)
	Contour Stellar Image	Tang et al. [47]	AlexNet & ACGAN	AWGN	BPSK, 4ASK, QPSK, OQPSK, 8PSK, 16QAM, 32QAM, 64QAM	[-6:2:14]	83.5% - 100%	/
	Contrast Enhanced Constellation Images & Regular Constellation Images	Huang et al. [48]	Compressive CNN (CCNN)	Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM	[-5:2:15]	46% - 100%	(NVIDIA GTX1080) Training Time: CCNN: 720s RNN: 810s DNN: 900s
	Grid Constellation Matrix	Huang et al. [49]	Contrastive Fully CNN (CFCN)	AWGN & Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM	[-5:5:15]	20% - 100%	(NVIDIA GTX1080) Training Time: CFCN: 10947s ResNet: 12795s DBN: 13169s SCAE: 16145s LSTM: 28681s
	Enhanced 3-Channel Constellation Diagram	Xie et al. [50]	CNN	AWGN	BPSK, QPSK, 8PSK	[-5:1:3]	Multi-label: 58% - 100% M ₂ M ₄ -aided: 52% - 100%	(NVIDIA GTX1080Ti) Inferring Time: Multi-label: 4.49ms M ₂ M ₄ -aided: 12.81ms
	Contour Stellar Image	Tu et al. [51]	Pruned AlexNet	AWGN	4ASK, BSPK, QPSK, OQPSK, 8PSK, 16QAM, 32QAM, 64QAM	[-6:2:6]	82% - 100%	(NVIDIA Jetson TX2) Inferring Time: Original: 19.416ms Pruned: 6.999ms
	Constellation Density Matrix	Kumar et al. [52]	ResNet-50 & Inception ResNet V2	AWGN	2ASK, 4ASK, BSPK, QPSK, 8PSK, 8QAM, 16QAM, 32QAM, 64QAM	[-5:5:15]	ResNet-50: 95.20% - 100% Inception ResNet V2: 16% - 100%	/
	Eye Diagram	Wang et al. [53]	Five-layer CNN	AWGN	RZ-OOK, NRZ-OOK, RZ-DPSK, 4PAM	[10:1:25] (OSNR)	95.20% - 100%	/
	Ambiguity Function Image	Dai et al. [54]	Stacke SAE-DNN	AWGN	ASK, PSK, QAM, FSK, MSK, LFM, OFDM	[-10:1:25]	90.40% - 99.8%	/
	Spectral Correlation Function Image	Mendis et al. [55]	DBN	AWGN	4FSK, 16QAM, BPSK, QPSK, OFDM	[-2:1:5]	89% - 100%	/
	Cyclic Correntropy Spectrum Graph	Ma et al. [57]	ResNet	AWGN & Non-Gaussian Noise	BPSK, 2ASK, 8ASK, 4QAM, 16QAM	[-5:5:10] (GSNR)	39% - 100%	/
	Bispectrum Graph	Li et al. [59]	AlexNet	AWGN	BPSK, 2ASK, 2FSK, 4FSK, 8FSK, LFM, OFDM	[-3,0,3,5,10]	15% - 97.7%	/

TABLE II
(Continued.) PERFORMANCE COMPARISON OF DIFFERENT DL-BASED MODULATION CLASSIFICATION ALGORITHMS

Signal Representation	Description	Reference	DL Network	Channel	Candidate Modulation	SNR (dB)	Accuracy	Complexity
Sequence	IQ Sequences	O'Shea et al. [7], [60]	Four-layer DNN	AWGN	BPSK, QPSK, 8PSK, 16QAM, 64QAM, BFSK, CPFSK, 4PAM, WB-FM, AM-SSB, AM-DSB	[-20:5:20]	15% - 87.4%	(NVIDIA Titan X) Inferring Time: 0.2s
	IQ Sequences	O'Shea et al. [8]	VGG-Net & ResNet	AWGN & Fading	BPSK, QPSK, OQPSK, 8PSK, 16PSK, 32PSK, 16QAM, 32QAM, 64QAM, 128QAM, 256QAM, OOK, 4ASK, 8ASK, 16APSK, 32APSK, 64APSK, 128APSK, AM-SSB-WC, AM-SSB-SC, AM-DSB-WC, AM-DSB-SC, FM, GMSK	[-20:5:20]	AWGN: ResNet: 4% - 76% VGG-Net: 4% - 63% Fading: ResNet: 5% - 79% VGG-Net: 5% - 68%	(NVIDIA V100) Training Time: ResNet: 14h
	IQ Sequences & Amplitude and Phase Sequences & FFT Sequences	Kulin et al. [61]	CNN	AWGN & Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM, CPFSK, GFSK, 4PAM, WB-FM, AM-DSB, AM-SSB	[-20:5:20]	IQ: 9% - 82% A/P: 10% - 85% FFT: 9% - 69%	/
	IQ Sequences	Haung et al. [63]	LSTM	AWGN & Fading	BPSK, QPSK, 8PSK, CPFSK, GFSK, 16QAM, 64QAM, 4PAM, WB-FM, AM-SSB, AM-DSB	[-20:2:18]	10% - 94%	/
	IQ Sequences	Liu et al. [66]	Seven-layer CNN & DenseNet & CLDNN & ResNet	AWGN & Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM, BFSK, CPFSK, 4PAM, WB-FM, AM-SSB, AM-DSB	[-25:5:30]	CNN: 10% - 83.8% DenseNet: 10% - 86.6% ResNet: 11% - 83.5% CLDNN: 10% - 88.5%	(NVIDIA M60) Training Time: DenseNet: 70h ResNet: 20h CLDNN: 50h
	IQ Sequences & Fourth-order Statistics	Zhang et al. [68]	Four-layer CNN & LSTM	AWGN & Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM, BFSK, CPFSK, 4PAM, WB-FM, AM-SSB, AM-DSB	[-20:2:18]	CNN: 12% - 97% LSTM: 12% - 98%	/
	IQ Sequences	Ma et al. [69]	Hybrid DNN (HDNN) & Layered ResNet (LRN)	AWGN & Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM, BFSK, CPFSK, 4PAM, WB-FM, AM-SSB, AM-DSB	[-20:2:20]	10% - 92%	Inferring Time: 2.59ms
	IQ Sequences	Yashashwi et al. [70]	CM+CNN	AWGN & Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM, BFSK, CPFSK, 4PAM, WB-FM, AM-SSB, AM-DSB	[-20:2:18]	10% - 90%	/
	IQ Sequences	Hermawan et al. [71]	IC-AMCNet	AWGN & Fading	RML2016.10a: BPSK, QPSK, 8PSK, 16QAM, 64QAM, BFSK, CPFSK, 4PAM, WB-FM, AM-SSB, AM-DSB RML2016.10b: BPSK, QPSK, 8PSK, AM-DSB, GFSK, CPFSK, 4PAM, 16QAM, 64QAM, WB-FM	[-20:2:18]	RadioML2016.10a: 10% - 83.4% RadioML2016.10b: 10% - 91.7%	(NVIDIA 1xTesla K80) Inferring Time: 0.0048ms
	IQ Sequences	Ali and Fan [72]	Unsorted DNN (UDNN)	AWGN & Fading	QPSK, 16PSK, 16QAM, 128QAM	[0:5:15]	AWGN: 85% - 100% Fading: 92% - 98.5% Phase Offset: 71% - 100% Frequency Offset: 80% - 97.5%	Overall Complexity: $O(N_W)$ (N_W is the number of weights)
	IQ Sequences & Amplitude and Phase Sequences & High-order Cumulants	Hong et al. [73]	CNN	AWGN & Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM	[0:5:30]	40% - 92%	/
	IQ Sequences	Shi et al. [74]	Five-layer CNN	AWGN	Dataset1: BPSK, QPSK, 8PSK Dataset2: BPSK, QPSK, 8PSK, 16QAM	[-10:5:20]	Dataset1: 65% - 100% Dataset2: 48% - 100%	/
	IQ Sequences	Meng et al. [75]	CNN	AWGN & Fading	BPSK, QPSK, 8PSK, 16APSK, 32APSK, 16QAM, 64QAM	[-6:2:10]	Coherent: 30% - 96% Incoherent: 50% - 100%	(NVIDIA GTX960) Inferring Time: 0.0775ms
	IQ Sequences	Zheng et al. [76]	CNN & ResNet	AWGN	BPSK, QPSK, 8PSK, OQPSK, 16QAM, 32QAM, 64QAM, 2FSK, 4FSK, 8FSK, 4PAM, 8PAM	[-20:2:30]	CNN: 10% - 92% CNN+Fusion: 10% - 97% ResNet: 10% - 99% ResNet+Fusion: 10% - 100%	/
	IQ Sequences (LOS & NLOS Signals)	Gu et al. [77]	CNN	Fading	2FSK, DQPSK, 16QAM, 4PAM, MSK, GMSK	[0:2:12]	83% - 98%	/
	IQ Sequences	Hu et al. [78]	RNN	AWGN & Fading & Uncertain Noise	BPSK, QPSK, 8PSK, 16QAM	[0:2:20]	AWGN: 61% - 98% WNGN: 59% - 98% TCNGN: 60% - 98% Frequency Offset: 72% - 82%	Inferring Time: AWGN: 9.2ms WNGN: 11.6ms TCNGN: 9.5ms
	IQ Sequences	Wang et al. [79]	Five-layer CNN	Flat Fading and Time-invariant Channel	BPSK, QPSK, 8PSK, 16QAM	[-10:5:10]	$N_F = 4$: $N_I = 1$: 49% - 100% $N_I = 2$: 42% - 99% $N_I = 4$: 45% - 96%	/

TABLE II
(Continued.) PERFORMANCE COMPARISON OF DIFFERENT DL-BASED MODULATION CLASSIFICATION ALGORITHMS

Signal Representation	Description	Reference	DL Network	Channel	Candidate Modulation	SNR (dB)	Accuracy	Complexity
Sequence	IQ Sequences & Amplitude and Phase Sequences	Rajendran et al. [80]	LSTM	AWGN & Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM, CPFSK, GFSK, 4PAM, WBFM, AM-DSB, AM-SSB	[-20:5:20]	RadioML: 10% - 92% Modified RadioML: 10% - 78%	(NVIDIA GTX980 Ti) Model Size: Memory Full Precision: 6.4MB TW_4BA: 400.15KB TW_FA: 400.15KB
	FFT Sequences & IQ Sequences	Mossad et al. [81]	Four-layer CNN	AWGN & Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM, CPFSK, GFSK, 4PAM, WBFM, AM-DSB	[-20:5:20]	11% - 86.97%	/
	Amplitude Histogram Sequences	Khan et al. [83], [84]	DNN	Fiber Optic	QPSK, 16QAM, 64QAM	[0:5:20] (OSNR)	Fiber Optic: 100% Fiber Nonlinearity: 99% - 100%	/
Combined	Fourth-order Cumulants & IQ Sequences	Zhang et al. [68]	Four-layer CNN & LSTM	AWGN & Fading	BPSK, QPSK, 8PSK, 16QAM, 64QAM, BFSK, CPFSK, 4PAM, WB-FM, AM-SSB, AM-DSB	[-20:2:18]	LSTM: 13% - 88% CNN: 12% - 87%	/
	IQ Sequences & Constellation Diagram	Wang et al. [85]	DrCNN & CNN	AWGN	BPSK, QPSK, 8PSK, GFSK, CPFSK, 4PAM, 16QAM, 64QAM	[-8:2:18]	43% - 98%	Parameters: 1,101,448
	Cyclic Spectra Image & Constellation Diagram	Wu et al. [86]	CNN	AWGN	BPSK, QPSK, 8PSK, 16QAM, 64QAM, CPFSK, GFSK, 4PAM, WB-FM, AM-SSB, AM-DSB	[-20:2:18]	16% - 92%	(NVIDIA GTX 1060) Parameters: 12k
	Joint Time-Frequency Image & Instantaneous Autocorrelation Image	Wang et al. [87]	Seven-layer CNN	AWGN	LFM, SF, BPSK, QPSK, 2FSK, 4FSK	[6:2:14]	97.1% - 99.8%	/
	SPWVD & BJD Images	Zhang et al. [88]	ResNet-152	AWGN	BPSK, QPSK, 2FSK, 4FSK, 2ASK, 4ASK, 16QAM, 64QAM, OFDM	[-4:2:8]	90% - 98.5%	Inferring Time: 1ms
	IQ Sequences & DOST Sequences	Hiremath et al. [90]	Eight-layer CNN	AWGN & Fading	BPSK, QPSK, 8PSK, GMSK, 16APSK, 64APSK, 16QAM, 64QAM, AM-DSB -WC, FM	[-8:2:8]	50% - 97.3%	/

transformations, e.g., AP sequences, can be easily derived with little calculation. For other sequences, such as FFT and histogram sequences, some extra calculation is needed, but the received signal can be compressed if the sequence length is reduced. The major challenge of modulation classification with sequence representation lies in the design of DNNs according to the characteristics of the sequences. If the networks are not well designed, the training process of DL may hardly converge.

The combined representation uses the combination of multiple features, images, or sequences to represent the received signal. It is able to integrate the advantages of multiple representations and obtain better classification performance. To achieve this goal, multiple representations should be as uncorrelated as possible. Moreover, they should be carefully combined based on a fusion rule. Otherwise, the defects of multiple representations may be accumulated and play a dominant role, which weakens the classification performance instead.

Although various DL-based modulation classification algorithms have represented the received signal in various formats, it is hard to compare their performance fairly as they utilize different DL networks, suffer from different channel impairments, consider different modulation sets, and concentrate on different SNR regions. According to Table II, almost all algorithms are able to achieve 100% accuracy, but the conditions to obtain this performance are ideal and strict, e.g., at high SNR larger than 10 dB, within a small candidate modulation set excluding analog modulation schemes, over an AWGN

channel without any fading and interference, and considering perfect synchronization and equalization with little frequency and phase offsets. Considering the practical datasets, such as RadioML that involves in analog modulation, multipath fading, center frequency offset, sample rate offset, and so on, the state-of-the-art algorithm exceeds 90% at 5 dB SNR, which is much better than traditional non-DL algorithms and demonstrates the superiority of DL-based modulation classification. Furthermore, deeper networks and higher SNRs lead to better classification performance.

In practical communications systems, some devices, such as mobile edge devices, do not have powerful computation capacity and large storage. To implement DL-based modulation classification in these devices, it is necessary to pay great attention to the complexity issue. Table II presents the complexity of existing algorithms in terms of either training time, inferring time, or model size. As shown in this table, most algorithms can be completed rapidly, which is critical for the deployment of these algorithms in practice.

VIII. CONCLUSION AND FUTURE DIRECTIONS

This article has surveyed the recent advancements in DL-based modulation classification from the perspective of signal representation and data preprocessing. Classification algorithms are categorized into four groups according to the techniques of signal representation. In feature representation, the received signal is converted into cumulants, spectral, or circular features. In image representation, it is transformed into constellation diagram, eye diagram, FP image, AF image, SCF

image, log-CCES graph, or ASB graph. In sequence representation, it is represented by IQ sequences, AP sequences, FFT sequences, or amplitude histogram sequences. Moreover, features, images, and sequences can be combined together for the representation of the received signal. All the representations are effective under the AWGN scenarios. Some of them, e.g., feature representation and image representation, have been shown to be good at combating the non-AWGN impairments if the features and images are appropriately designed and selected.

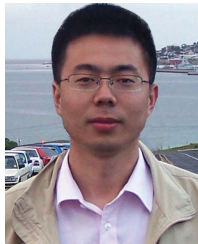
Note that different representations retain different characteristics of the received signal, and the characteristic suitable for distinguishing one modulation scheme may be useless for recognizing another one. Consequently, it will be of great value to study the impacts of representations on modulation schemes, with which proper representations could be determined according to the candidate modulation schemes. In addition, since different DL networks are good at handling different formats of data, considering each representation, an appropriate network structure should be selected or designed to fully learn from the signal representation, which also needs further investigation. Besides, the combined representation is superior in exploiting multiple-signal representations of features, images, and sequences, while how to fuse these representations are still not well researched. One promising solution may be to take multimodal DL into consideration and implement either data, feature, or decision fusion to achieve high-accuracy and low-complexity modulation classification. Meanwhile, these methods and techniques may also be applied in non-DL algorithms, e.g., SVM- or KNN-based modulation classification, which are worthy to be studied further in the future. Finally, with the evolution of network structure and computation hardware, DL-based modulation classification can be broadly applied in future communications systems, and all techniques of signal representation and data preprocessing summarized in this article may provide valuable experiences for the implementation of DL to address other communications tasks, such as full wireless environment awareness, intelligent signal processing, and smart communications system design. Furthermore, since the mobile environment is dynamic, some wireless parameters may change, causing the mismatch between the training data and inferring data, which inevitably impairs the classification performance and is worthy of further research.

REFERENCES

- [1] Y. Zou, Y.-D. Yao, and B. Zheng, "Cooperative relay techniques for cognitive radio systems: Spectrum sensing and secondary user transmissions," *IEEE Commun. Mag.*, vol. 50, no. 4, pp. 98–103, Apr. 2012.
- [2] H. Wang and Y.-D. Yao, "Primary user boundary detection in cognitive radio networks: Estimated secondary user locations and impact of malicious secondary users," *IEEE Trans. Veh. Technol.*, vol. 67, no. 5, pp. 4577–4588, May 2018.
- [3] M. Alymani, M. H. Alhazmi, A. Almarhabi, H. Alhazmi, A. Samarkandi, and Y.-D. Yao, "Rician K-factor estimation using deep learning," in *Proc. 29th Wireless Opt. Commun. Conf. (WOCC)*, May 2020, pp. 1–4.
- [4] M. He, S. Peng, H. Wang, and Y.-D. Yao, "Identification of ISM band signals using deep learning," in *Proc. 29th Wireless Opt. Commun. Conf. (WOCC)*, May 2020, pp. 1–4.
- [5] X. Qin, S. Peng, X. Yang, and Y.-D. Yao, "Deep learning based channel code recognition using TextCNN," in *Proc. IEEE Int. Symp. Dyn. Spectr. Access Netw. (DySPAN)*, Nov. 2019, pp. 1–5.
- [6] O. A. Dobre, A. Abdi, Y. Bar-Ness, and W. Su, "Survey of automatic modulation classification techniques: Classical approaches and new trends," *IET Commun.*, vol. 1, no. 2, pp. 137–156, Apr. 2007.
- [7] T. J. O'Shea, J. Corgan, and T. C. Clancy, "Convolutional radio modulation recognition networks," in *Proc. 17th Int. Conf. Eng. Appl. Neural Netw. (EANN)*, 2016, pp. 213–226.
- [8] T. J. O'Shea, T. Roy, and T. C. Clancy, "Over-the-air deep learning based radio signal classification," *IEEE J. Sel. Topics Signal Process.*, vol. 12, no. 1, pp. 168–179, Feb. 2018.
- [9] S. Peng, H. Jiang, H. Wang, H. Alwageed, and Y.-D. Yao, "Modulation classification using convolutional neural network based deep learning model," in *Proc. 26th Wireless Opt. Commun. Conf. (WOCC)*, Apr. 2017, pp. 1–5.
- [10] S. Peng *et al.*, "Modulation classification based on signal constellation diagrams and deep learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 3, pp. 718–727, Mar. 2019.
- [11] S. Hu, Y.-D. Yao, and Z. Yang, "MAC protocol identification approach for implement smart cognitive radio," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2012, pp. 5608–5612.
- [12] S. Hu, Y.-D. Yao, and Z. Yang, "MAC protocol identification using support vector machines for cognitive radio networks," *IEEE Wireless Commun.*, vol. 21, no. 1, pp. 52–60, Feb. 2014.
- [13] Y. Zhou, S. Peng, and Y. Yao, "MAC protocol identification using convolutional neural networks," in *Proc. 29th Wireless Opt. Commun. Conf. (WOCC)*, May 2020, pp. 1–4.
- [14] N. Hu and Y.-D. Yao, "Identification of legacy radios in a cognitive radio network using a radio frequency fingerprinting based method," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2012, pp. 1597–1602.
- [15] K. Alshathri, H. Xia, V. Lawrence, and Y.-D. Yao, "Cellular system identification using deep learning: GSM, UMTS and LTE," in *Proc. 28th Wireless Opt. Commun. Conf. (WOCC)*, May 2019, pp. 1–4.
- [16] H. Xia *et al.*, "Cellular signal identification using convolutional neural networks: AWGN and Rayleigh fading channels," in *Proc. IEEE Int. Symp. Dyn. Spectr. Access Netw. (DySPAN)*, Nov. 2019, pp. 1–5.
- [17] M. H. Alhazmi, M. Alymani, H. Alhazmi, A. Almarhabi, A. Samarkandi, and Y.-D. Yao, "5G signal identification using deep learning," in *Proc. 29th Wireless Opt. Commun. Conf. (WOCC)*, May 2020, pp. 1–5.
- [18] J. L. Xu, W. Su, and M. Zhou, "Likelihood-ratio approaches to automatic modulation classification," *IEEE Trans. Syst., Man, Cybern. C, Appl. Rev.*, vol. 41, no. 4, pp. 455–469, Jul. 2011.
- [19] Q. Shi and Y. Karasawa, "Automatic modulation identification based on the probability density function of signal phase," *IEEE Trans. Commun.*, vol. 60, no. 4, pp. 1033–1044, Apr. 2012.
- [20] H.-C. Wu, M. Saquib, and Z. Yun, "Novel automatic modulation classification using cumulant features for communications via multipath channels," *IEEE Trans. Wireless Commun.*, vol. 7, no. 8, pp. 3098–3105, Aug. 2008.
- [21] B. Ramkumar, "Automatic modulation classification for cognitive radios using cyclic feature detection," *IEEE Circuits Syst. Mag.*, vol. 9, no. 2, pp. 27–45, Jun. 2009.
- [22] S. U. Pawar and J. F. Doherty, "Modulation recognition in continuous phase modulation using approximate entropy," *IEEE Trans. Inf. Forensics Security*, vol. 6, no. 3, pp. 843–852, Sep. 2011.
- [23] Y. Deng and Z. Wang, "Modulation recognition of MAPSK signals using template matching," *Electron. Lett.*, vol. 50, no. 25, pp. 1986–1988, Dec. 2014.
- [24] L. Xie and Q. Wan, "Automatic modulation recognition for phase shift keying signals with compressive measurements," *IEEE Wireless Commun. Lett.*, vol. 7, no. 2, pp. 194–197, Apr. 2018.
- [25] S. Kharbech, I. Dayoub, M. Zwingelstein-Colin, and E. P. Simon, "On classifiers for blind feature-based automatic modulation classification over multiple-input-multiple-output channels," *IET Commun.*, vol. 10, no. 7, pp. 790–795, May 2016.
- [26] W. B. Chikha, I. Dayoub, W. Hamouda, and R. Attia, "Modulation recognition for MIMO relaying broadcast channels with direct link," *IEEE Wireless Commun. Lett.*, vol. 3, no. 1, pp. 50–53, Feb. 2014.
- [27] M. W. Aslam, Z. Zhu, and A. K. Nandi, "Automatic modulation classification using combination of genetic programming and KNN," *IEEE Trans. Wireless Commun.*, vol. 11, no. 8, pp. 2742–2750, Aug. 2012.
- [28] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

- [29] Q. Mao, F. Hu, and Q. Hao, "Deep learning for intelligent wireless networks: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2595–2621, Jun. 2018.
- [30] A. Swami and B. M. Sadler, "Hierarchical digital modulation classification using cumulants," *IEEE Trans. Commun.*, vol. 48, no. 3, pp. 416–429, Mar. 2000.
- [31] J. Lee, B. Kim, J. Kim, D. Yoon, and J. W. Choi, "Deep neural network-based blind modulation classification for fading channels," in *Proc. Int. Conf. Inf. Commun. Technol. Converg. (ICTC)*, Oct. 2017, pp. 551–554.
- [32] A. Ali and F. Yangyu, "Automatic modulation classification using deep learning based on sparse autoencoders with nonnegativity constraints," *IEEE Signal Process. Lett.*, vol. 24, no. 11, pp. 1626–1630, Nov. 2017.
- [33] W. Xie, S. Hu, C. Yu, P. Zhu, X. Peng, and J. Ouyang, "Deep learning in digital modulation recognition using high order cumulants," *IEEE Access*, vol. 7, pp. 63760–63766, 2019.
- [34] S. H. Lee, K.-Y. Kim, J. H. Kim, and Y. Shin, "Effective feature-based automatic modulation classification method using DNN algorithm," in *Proc. Int. Conf. Artif. Intell. Inf. Commun. (ICAIC)*, Feb. 2019, pp. 557–559.
- [35] Z. Zhu and A. K. Nandi, *Autom. Modulation Classification: Principles, Algorithms and Applications*. London, U.K.: Wiley, 2015.
- [36] M. H. Shah and X. Dang, "Classification of spectrally efficient constant envelope modulations based on radial basis function network and deep learning," *IEEE Commun. Lett.*, vol. 23, no. 9, pp. 1529–1533, Sep. 2019.
- [37] W. Shi, D. Liu, X. Cheng, Y. Li, and Y. Zhao, "Particle swarm optimization-based deep neural network for digital modulation recognition," *IEEE Access*, vol. 7, pp. 104591–104600, 2019.
- [38] E. E. Azzouz and A. K. Nandi, "Automatic identification of digital modulation types," *Signal Process.*, vol. 47, no. 1, pp. 55–69, Nov. 1995.
- [39] A. K. Nandi and E. E. Azzouz, "Algorithms for automatic modulation recognition of communication signals," *IEEE Trans. Commun.*, vol. 46, no. 4, pp. 431–436, Apr. 1998.
- [40] J. H. Lee, K.-Y. Kim, and Y. Shin, "Feature image-based automatic modulation classification method using CNN algorithm," in *Proc. Int. Conf. Artif. Intell. Inf. Commun. (ICAIC)*, Feb. 2019, pp. 1–4.
- [41] K. L. Davidson, J. R. Goldschneider, L. Cazzanti, and J. W. Pitton, "Feature-based modulation classification using circular statistics," in *Proc. Mil. Commun. Conf. (IEEE MILCOM)*, vol. 2, Oct. 2004, pp. 765–771.
- [42] K. C. Ho, W. Prokopiw, and Y. T. Chan, "Modulation identification of digital signals by the wavelet transform," *IEE Proc.-Radar, Sonar Navigat.*, vol. 147, no. 4, pp. 169–176, Aug. 2000.
- [43] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
- [44] F. Wang, Y. Wang, and X. Chen, "Graphic constellations and DBN based automatic modulation classification," in *Proc. IEEE 85th Veh. Technol. Conf. (VTC Spring)*, Jun. 2017, pp. 1–5.
- [45] G. E. Hinton, S. Osindero, and Y.-W. Teh, "A fast learning algorithm for deep belief nets," *Neural Comput.*, vol. 18, no. 7, pp. 1527–1554, Jul. 2006.
- [46] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.
- [47] B. Tang, Y. Tu, Z. Zhang, and Y. Lin, "Digital signal modulation classification with data augmentation using generative adversarial nets in cognitive radio networks," *IEEE Access*, vol. 6, pp. 15713–15722, 2018.
- [48] S. Huang *et al.*, "Automatic modulation classification using compressive convolutional neural network," *IEEE Access*, vol. 7, pp. 79636–79643, 2019.
- [49] S. Huang, Y. Jiang, Y. Gao, Z. Feng, and P. Zhang, "Automatic modulation classification using contrastive fully convolutional network," *IEEE Wireless Commun. Lett.*, vol. 8, no. 4, pp. 1044–1047, Aug. 2019.
- [50] X. Xie, Y. Ni, S. Peng, and Y.-D. Yao, "Deep learning based automatic modulation classification for varying SNR environment," in *Proc. 28th Wireless Opt. Commun. Conf. (WOCC)*, May 2019, pp. 1–5.
- [51] Y. Tu and Y. Lin, "Deep neural network compression technique towards efficient digital signal modulation recognition in edge device," *IEEE Access*, vol. 7, pp. 58113–58119, 2019.
- [52] Y. Kumar, M. Sheoran, G. Jajoo, and S. K. Yadav, "Automatic modulation classification based on constellation density using deep learning," *IEEE Commun. Lett.*, vol. 24, no. 6, pp. 1275–1278, Jun. 2020.
- [53] D. Wang *et al.*, "Modulation format recognition and OSNR estimation using CNN-based deep learning," *IEEE Photon. Technol. Lett.*, vol. 29, no. 19, pp. 1667–1670, Oct. 1, 2017.
- [54] A. Dai, H. Zhang, and H. Sun, "Automatic modulation classification using stacked sparse auto-encoders," in *Proc. IEEE 13th Int. Conf. Signal Process. (ICSP)*, Nov. 2016, pp. 248–252.
- [55] G. J. Mendis, J. Wei, and A. Madanayake, "Deep learning-based automated modulation classification for cognitive radio," in *Proc. IEEE Int. Conf. Commun. Syst. (ICCS)*, Dec. 2016, pp. 1–6.
- [56] G. J. Mendis, J. Wei-Kocsis, and A. Madanayake, "Deep learning based radio-signal identification with hardware design," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 5, pp. 2516–2531, Oct. 2019.
- [57] J. Ma, S.-C. Lin, H. Gao, and T. Qiu, "Automatic modulation classification under non-Gaussian noise: A deep residual learning approach," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2019, pp. 1–6.
- [58] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [59] Y. Li, G. Shao, and B. Wang, "Automatic modulation classification based on bispectrum and CNN," in *Proc. IEEE 8th Joint Int. Inf. Technol. Artif. Intell. Conf. (ITAIC)*, May 2019, pp. 311–316.
- [60] T. J. O'Shea and N. West, "Radio machine learning dataset generation with GNU radio," in *Proc. GNU Radio Conf.*, 2016, pp. 1–6.
- [61] M. Kulin, T. Kazaz, I. Moerman, and E. De Poorter, "End-to-end learning from spectrum data: A deep learning approach for wireless signal identification in spectrum monitoring applications," *IEEE Access*, vol. 6, pp. 18484–18501, 2018.
- [62] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2015, *arXiv:1409.1556*. [Online]. Available: <https://arxiv.org/abs/1409.1556>
- [63] L. Huang, W. Pan, Y. Zhang, L. Qian, N. Gao, and Y. Wu, "Data augmentation for deep learning-based radio modulation classification," *IEEE Access*, vol. 8, pp. 1498–1506, 2020.
- [64] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.
- [65] T. N. Sainath, O. Vinyals, A. Senior, and H. Sak, "Convolutional, long short-term memory, fully connected deep neural networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2015, pp. 4580–4584.
- [66] X. Liu, D. Yang, and A. E. Gamal, "Deep neural network architectures for modulation classification," in *Proc. 51st Asilomar Conf. Signals, Syst., Comput.*, Oct. 2017, pp. 915–919.
- [67] K. Greff, R. K. Srivastava, J. Koutnik, B. R. Steunebrink, and J. Schmidhuber, "LSTM: A search space odyssey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 10, pp. 2222–2232, Oct. 2017.
- [68] M. Zhang, Y. Zeng, Z. Han, and Y. Gong, "Automatic modulation recognition using deep learning architectures," in *Proc. IEEE 19th Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, Jun. 2018, pp. 1–5.
- [69] H. Ma *et al.*, "Cross model deep learning scheme for automatic modulation classification," *IEEE Access*, vol. 8, pp. 78923–78931, 2020.
- [70] K. Yashashwi, A. Sethi, and P. Chaporkar, "A learnable distortion correction module for modulation recognition," *IEEE Wireless Commun. Lett.*, vol. 8, no. 1, pp. 77–80, Feb. 2019.
- [71] A. P. Hermawan, R. R. Ginanjar, D.-S. Kim, and J.-M. Lee, "CNN-based automatic modulation classification for beyond 5G communications," *IEEE Commun. Lett.*, vol. 24, no. 5, pp. 1038–1041, May 2020.
- [72] A. Ali and F. Yangyu, "k-sparse autoencoder-based automatic modulation classification with low complexity," *IEEE Commun. Lett.*, vol. 21, no. 10, pp. 2162–2165, Oct. 2017.
- [73] S. Hong, Y. Zhang, Y. Wang, H. Gu, G. Gui, and H. Sari, "Deep learning-based signal modulation identification in OFDM systems," *IEEE Access*, vol. 7, pp. 114631–114638, 2019.
- [74] J. Shi, S. Hong, C. Cai, Y. Wang, H. Huang, and G. Gui, "Deep learning-based automatic modulation recognition method in the presence of phase offset," *IEEE Access*, vol. 8, pp. 42841–42847, 2020.
- [75] F. Meng, P. Chen, L. Wu, and X. Wang, "Automatic modulation classification: A deep learning enabled approach," *IEEE Trans. Veh. Technol.*, vol. 67, no. 11, pp. 10760–10772, Nov. 2018.
- [76] S. Zheng, P. Qi, S. Chen, and X. Yang, "Fusion methods for CNN-based automatic modulation classification," *IEEE Access*, vol. 7, pp. 66496–66504, 2019.
- [77] H. Gu, Y. Wang, S. Hong, and G. Gui, "Blind channel identification aided generalized automatic modulation recognition based on deep learning," *IEEE Access*, vol. 7, pp. 110722–110729, 2019.
- [78] S. Hu, Y. Pei, P. P. Liang, and Y.-C. Liang, "Deep neural network for robust modulation classification under uncertain noise conditions," *IEEE Trans. Veh. Technol.*, vol. 69, no. 1, pp. 564–577, Jan. 2020.

- [79] Y. Wang, J. Wang, W. Zhang, J. Yang, and G. Gui, "Deep learning-based cooperative automatic modulation classification method for MIMO systems," *IEEE Trans. Veh. Technol.*, vol. 69, no. 4, pp. 4575–4579, Apr. 2020.
- [80] S. Rajendran, W. Meert, D. Giustiniano, V. Lenders, and S. Pollin, "Deep learning models for wireless signal classification with distributed low-cost spectrum sensors," *IEEE Trans. Cognit. Commun. Netw.*, vol. 4, no. 3, pp. 433–445, Sep. 2018.
- [81] O. S. Mossad, M. ElNainay, and M. Torki, "Deep convolutional neural network with multi-task learning scheme for modulations recognition," in *Proc. 15th Int. Wireless Commun. Mobile Comput. Conf. (IWCMC)*, Jun. 2019, pp. 1644–1649.
- [82] S. Rajendran *et al.*, "Electrosense: Open and big spectrum data," *IEEE Commun. Mag.*, vol. 56, no. 1, pp. 210–217, Jan. 2018.
- [83] F. N. Khan, K. Zhong, W. H. Al-Arashi, C. Yu, C. Lu, and A. P. T. Lau, "Modulation format identification in coherent receivers using deep machine learning," *IEEE Photon. Technol. Lett.*, vol. 28, no. 17, pp. 1886–1889, Sep. 1, 2016.
- [84] F. N. Khan *et al.*, "Joint OSNR monitoring and modulation format identification in digital coherent receivers using deep neural networks," *Opt. Exp.*, vol. 25, no. 15, pp. 17767–17776, Jul. 2017.
- [85] Y. Wang, M. Liu, J. Yang, and G. Gui, "Data-driven deep learning for automatic modulation recognition in cognitive radios," *IEEE Trans. Veh. Technol.*, vol. 68, no. 4, pp. 4074–4077, Apr. 2019.
- [86] H. Wu, Y. Li, L. Zhou, and J. Meng, "Convolutional neural network and multi-feature fusion for automatic modulation classification," *Electron. Lett.*, vol. 55, no. 16, pp. 895–897, Aug. 2019.
- [87] F. Wang, C. Yang, S. Huang, and H. Wang, "Automatic modulation classification based on joint feature map and convolutional neural network," *IET Radar, Sonar Navigat.*, vol. 13, no. 6, pp. 998–1003, Jun. 2019.
- [88] Z. Zhang, C. Wang, C. Gan, S. Sun, and M. Wang, "Automatic modulation classification using convolutional neural network with features fusion of SPWVD and BJD," *IEEE Trans. Signal Inf. Process. Netw.*, vol. 5, no. 3, pp. 469–478, Sep. 2019.
- [89] R. G. Stockwell, "A basis for efficient representation of the S-transform," *Digit. Signal Process.*, vol. 17, no. 1, pp. 371–393, Jan. 2007.
- [90] S. M. Hiremath, S. Behura, S. Kedia, S. Deshmukh, and S. K. Patra, "Deep learning-based modulation classification using time and stockwell domain channeling," in *Proc. Nat. Conf. Commun. (NCC)*, Feb. 2019, pp. 1–6.



Shengliang Peng (Member, IEEE) received the Ph.D. degree in electric engineering from the Nation Mobile Communications Laboratory, Southeast University, Nanjing, China, in 2011.

Since 2011, he has been with the School of Information Science and Engineering, Huaqiao University, Xiamen, China, where he is currently an Associate Professor. From 2016 to 2017, he was a Visiting Research Scholar with the Department of Electrical and Computer Engineering, Stevens Institute of Technology, Hoboken, NJ, USA. His

current research interests include cognitive radio, the Internet of Things, and deep learning.



Shujun Sun received the B.Eng. degree in information engineering from Huaqiao University, Quanzhou, China, in 2019, where she is currently pursuing the M.Eng. degree in information and communication engineering with Huaqiao University, Xiamen, China.

Her current research interests include wireless communications and artificial intelligence.



Yu-Dong Yao (Fellow, IEEE) received the B.Eng. and M.Eng. degrees in electrical engineering from the Nanjing University of Posts and Telecommunications, Nanjing, China, in 1982 and 1985, respectively, the Ph.D. degree in electrical engineering from Southeast University, Nanjing, in 1988, and the M.E. degree (Honoris Causa) from the Stevens Institute of Technology, Hoboken, NJ, USA, in 2018.

He was a Visiting Student with Carleton University, Ottawa, ON, Canada, from 1987 to 1988. From 1989 to 2000, he was with Carleton University, Spar Aerospace Ltd., Montreal, QC, Canada, and Qualcomm Inc., San Diego, CA, USA. He has been with the Stevens Institute of Technology, since 2000. He was the Chair of the Electrical and Computer Engineering Department, Stevens Institute of Technology, from 2007 to 2018. He is also the Director of the Stevens Wireless Information Systems Engineering Laboratory, Stevens Institute of Technology. He holds one Chinese patent and 13 U.S. patents. His current research interests include wireless communications, cognitive radio, machine learning, and deep learning.

Dr. Yao was elected as a fellow of the National Academy of Inventors in 2015 and the Canadian Academy of Engineering in 2017 for his contributions to wireless communications systems. He was an Associate Editor of the IEEE COMMUNICATIONS LETTERS from 2000 to 2008 and the IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY from 2001 to 2006 and an Editor of the IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS from 2001 to 2005.